

MICROLOCALIZATION AND SINGULAR SUPPORTS OF CONSTRUCTIBLE ÉTALE SHEAVES

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ABSTRACT. Let X be a smooth scheme over a perfect field of characteristic $p > 0$, and let F be a constructible complex of finite Tor-dimension with finite coefficients of characteristic prime to p . We prove

$$\mathrm{SS}_\mu(F) \subseteq \mathrm{SS}(F),$$

where $\mathrm{SS}_\mu(F)$ is Saito's microlocal singular support and $\mathrm{SS}(F)$ is Beilinson's singular support. This answers a question of Saito. For perverse ℓ -adic sheaves, the inclusion is an equality.

As applications, we resolve another question of Saito regarding support estimates for microlocalization along a smooth closed subscheme, and we prove Saito's conjecture on characteristic classes (*Invent. Math.* 207: 597-695, 2017), showing that the cohomological characteristic classes of Abbes and Saito are the cycle classes associated with the corresponding characteristic cycles.

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1. INTRODUCTION

1.1. Let k be a perfect field of characteristic $p > 0$, and let Λ be a finite local ring of residue characteristic $\ell \neq p$. Let X be a smooth k -scheme, and let $F \in D_{\mathrm{ctf}}^b(X, \Lambda)$. The singular support $\mathrm{SS}(F)$ defined by Beilinson [Bei16] is a closed conical subset of the cotangent bundle T^*X , characterized

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via universal local acyclicity of test pairs. Furthermore, Saito [Sai17] constructs the characteristic cycle $\text{CC}(F)$ supported on $\text{SS}(F)$, which is characterized by the Milnor formula.

Following the approach of Kashiwara and Schapira [KS90] in the transcendental setting, Saito [Sai26] introduces microlocal analogues of singular support and characteristic cycle for $F \in D_{\text{ctf}}^b(X, \Lambda)$. Let $\Delta : X \hookrightarrow X \times X$ be the diagonal and put

$$(1.1.1) \quad \mathcal{H}_F = R\mathcal{H}om(\text{pr}_1^* F, \text{pr}_2^! F).$$

We fix a nontrivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$. Verdier specialization $\nu_\Delta(\mathcal{H}_F)$ of $\mathcal{H}_F$ along the diagonal Δ yields a monodromic complex on the normal bundle $T_X(X \times X) \simeq TX$. More precisely, this complex is defined on $TX \times_k \eta_0$, where η_0 is the generic point of the henselization of $\mathbb{A}_k^1$ at 0. Applying the Fourier-Deligne transform $\mathcal{F}_\psi$ on TX , Saito sets

$$(1.1.2) \quad \mu\mathcal{H}om(F, F) = \mathcal{F}_\psi \nu_\Delta(\mathcal{H}_F),$$

and defines the microlocal singular support and microlocal characteristic cycle

$$\text{SS}_\mu(F) = \text{pr}_{T^*X} \text{supp} \mu\mathcal{H}om(F, F) \subseteq T^*X \quad \text{and} \quad \text{CC}_\mu(F) \in H_{\text{SS}_\mu(F)}^0(T^*X, \pi^* \mathcal{K}_{X/k}),$$

where $\pi : T^*X \rightarrow X$ is the structure map and $\mathcal{K}_{X/k} = Ra^! \Lambda$ with $a : X \rightarrow \text{Spec} k$ the structure morphism.

The relation between $\text{SS}(F)$ and $\text{SS}_\mu(F)$ is not immediate from their definitions. Beilinson's singular support $\text{SS}(F)$ is detected by functions and local acyclicity, whereas Saito's microlocal singular support $\text{SS}_\mu(F)$ is obtained by applying the Fourier-Deligne transform to Verdier specialization along the diagonal $\Delta : X \hookrightarrow X \times X$. Saito asks the following questions.

Question 1.2 (Saito, [Sai26, Question 2.1.9]). Let F be a constructible complex on a smooth k -scheme.

- (1) Do we have $\text{SS}_\mu(F) \subseteq \text{SS}(F)$?
- (2) Does $\text{CC}_\mu(F)$ equal the cycle class of $\text{CC}(F)$?

Our first main result answers both parts of Question 1.2 affirmatively.

Theorem 1.3 (Theorem 4.2, Theorem 8.2, Corollary 6.1 and Corollary 8.4). *Let X be a smooth connected k -scheme and $F \in D_{\text{ctf}}^b(X, \Lambda)$. We have*

$$(1.3.1) \quad \text{SS}_\mu(F) \subseteq \text{SS}(F),$$

$$(1.3.2) \quad \text{CC}_\mu(F) = \text{cl}(\text{CC}(F)),$$

where $\text{cl} : Z_{\dim(X)}(\text{SS}(F)) \rightarrow H_{\text{SS}(F)}^0(T^*X, \pi^* \mathcal{K}_{X/k})$ is the cycle class map.

Both assertions also hold for constructible complexes with coefficients in a finite extension $E/\mathbf{Q}_\ell$, interpreted on the pro-étale site.

When X is a smooth curve, Theorem 1.3 follows from [Sai26, Lemmas 2.1.2 and 2.1.5].

1.4. It is natural to ask when the inclusion $\text{SS}_\mu(F) \subseteq \text{SS}(F)$ is an equality. For a perverse sheaf, each irreducible component of the singular support can be detected by a suitable pencil. Combining this observation with Theorem 1.3, we obtain the following result.

Theorem 1.5 (Theorem 8.6). *Let E be a finite extension of $\mathbf{Q}_\ell$. For every perverse E -sheaf F on a smooth k -scheme, we have*

$$(1.5.1) \quad \text{SS}_\mu(F) = \text{SS}(F).$$

1.6. As a first application of Theorem 1.3, we obtain a support estimate for microlocalization along a smooth closed subscheme. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth k -schemes. Let $T_Z X$ be the normal bundle of $Z \hookrightarrow X$ and $\mathcal{F}_{\psi, T_Z X/Z}$ the Fourier-Deligne transform on $T_Z X \rightarrow Z$. We define

$$(1.6.1) \quad \mu_{Z/X}(F) = \mathcal{F}_{\psi, T_Z X/Z}(\nu_{Z/X} F) \in D_{\text{ctf}}^b(T_Z^* X, \Lambda),$$

where $\nu_{Z/X} F$ is the Verdier specialization of F along $Z \hookrightarrow X$. Saito asks whether the support of this microlocalization is contained in $\text{SS}(F) \cap T_Z^* X$ [Sai26, Question 2.5.3]. We construct a retraction

$$(1.6.2) \quad (q^\vee)^* \mu \mathcal{H}om(F, F) \otimes \mu_{Z/X}(F) \longrightarrow \mu_{Z/X}(F),$$

where $q^\vee$ is the dual of the quotient map $q : TX \times_X Z \twoheadrightarrow T_Z X$. We use this retraction to prove the following corollary, which gives an affirmative answer to [Sai26, Question 2.5.3].

Theorem 1.7 (Theorem 5.2). *For every closed immersion $Z \hookrightarrow X$ of smooth schemes and every $F \in D_{\text{ctf}}^b(X, \Lambda)$, we have*

$$(1.7.1) \quad \text{supp} \mu_{Z/X}(F) \subseteq \text{SS}_\mu(F) \cap T_Z^* X \subseteq \text{SS}(F) \cap T_Z^* X.$$

1.8. As a second application of Theorem 1.3, we prove Saito's conjecture on cohomological characteristic classes.

Conjecture 1.9 (Saito, [Sai17, Conjecture 6.8]). *Let Z be a k -scheme, and let $i : Z \rightarrow X$ be a closed immersion into a smooth k -scheme X . Let Λ be a finite local ring whose residue characteristic is invertible in k . For every $F \in D_{\text{ctf}}^b(Z, \Lambda)$, we have*

$$(1.9.1) \quad C_{Z/k}(F) = \text{cl}_Z(cc_{Z,0}(F)) \quad \text{in } H^0(Z, \mathcal{K}_{Z/k}),$$

where $cc_{Z,0}(F)$ denotes the characteristic class associated with the cycle $\text{CC}(i_* F)$ (see [Sai17, Definition 6.7]).

In [YZ25, Theorem 1.2], the second author and Zhao prove this conjecture in the case where Z is smooth and quasi-projective by using non-acyclicity classes and the fibration method. Using microlocalization, we prove Saito's conjecture in full generality.

Theorem 1.10 (Theorem 6.7). *Conjecture 1.9 holds.*

1.11. We outline the proof of Theorem 1.3 with finite coefficients. By a standard reduction, we may assume that $X = P = \mathbb{P}^n$ is a projective space of dimension $n \geq 1$. Let $P^\vee$ be the dual projective space and let

$$Q = \{(x, H) : x \in H\} \subset P \times P^\vee, \quad p : Q \longrightarrow P, \quad p^\vee : Q \longrightarrow P^\vee$$

be the incidence correspondence and its projections respectively. Beilinson's results on Radon transforms [Bei16, Theorem] identify the projectivization of $\text{SS}(F)$ with the non-ULA locus¹ of the morphism p with respect to $G = p^{\vee*} R p_*^\vee p^* F[2n-2](n-1)$. It therefore suffices to show that $\mu \mathcal{H}om(F, F)$ vanishes on the cotangent lines parametrized by the ULA locus.

We consider the base change of the diagonal $\Delta_P : P \hookrightarrow P \times P$ along $p \times p$

$$(1.11.1) \quad \begin{array}{ccc} Q \times_P Q & \hookrightarrow & Q \times Q \\ \downarrow & & \downarrow p \times p \\ P & \xrightarrow{\Delta_P} & P \times P. \end{array}$$

It induces a natural morphism $\Pi : T_{Q \times_P Q}(Q \times Q) \simeq Q \times_P Q \times_P TP \rightarrow T_P(P \times P) \simeq TP$. A key step is to construct the following retraction (see Lemma 3.5):

$$(1.11.2) \quad \nu_{\Delta_P}(\mathcal{H}_F) \longrightarrow R\Pi_* \nu_{Q \times_P Q/Q \times Q}(\mathcal{H}_G) \longrightarrow \nu_{\Delta_P}(\mathcal{H}_F).$$

¹This is the smallest closed subset of Q outside which p is universally locally acyclic relatively to G .

Applying the Fourier-Deligne transform to (1.11.2), we obtain $\mu\mathcal{H}om(F, F)$ as a retract of

$$(1.11.3) \quad \mathcal{F}_\psi R\Pi_* \nu_{Q \times_P Q/Q \times Q}(\mathcal{H}_G).$$

We use this retraction to estimate the support of $\mu\mathcal{H}om(F, F)$, and finally we prove (1.3.1) for finite coefficients. For E -coefficients, we work in the constructible derived category on the pro-étale site [BS15] and use Barrett's extension [Bar24] of Beilinson's results to the E -adic setting.

1.12. We also establish a Künneth formula for $\mu\mathcal{H}om$. Let X and Y be smooth k -schemes. For constructible complexes $F \in D_{\text{ctf}}^b(X, \Lambda)$ and $G \in D_{\text{ctf}}^b(Y, \Lambda)$, we construct a canonical isomorphism

$$(1.12.1) \quad \mu\mathcal{H}om(F \boxtimes G, F \boxtimes G) \simeq \mu\mathcal{H}om(F, F) \boxtimes \mu\mathcal{H}om(G, G).$$

This gives the product formulas

$$(1.12.2) \quad \text{SS}_\mu(F \boxtimes G) = \text{SS}_\mu(F) \times \text{SS}_\mu(G), \quad \text{CC}_\mu(F \boxtimes G) = \text{CC}_\mu(F) \boxtimes \text{CC}_\mu(G).$$

1.13. The paper is organized as follows. Section 2 develops the necessary preliminaries on deformation to the normal bundle and recalls the Verdier specialization functor. Section 3 establishes a retraction for Radon transforms and constructs (1.11.2). Section 4 proves (1.3.1) for finite coefficients, while Section 5 proves Theorem 1.7. Section 6 proves Theorem 1.10. Section 7 establishes a Künneth formula for $\mu\mathcal{H}om$. Finally, Section 8 establishes (1.3.2) and proves Theorem 1.5.

1.14. During 2016–2018, when the second author was a postdoctoral researcher at the University of Regensburg, Denis-Charles Cisinski suggested developing a theory of singular support and microlocalization in the motivic setting. While a notion of singular support for motives was successfully constructed, the difficulties arising from wild ramification led the second author to remain skeptical about the existence of a satisfactory theory of microlocalization. After completing the joint work with Zhao [YZ25], however, he came to believe that an appropriate form of microlocalization might exist, although no substantial progress was made at that stage. It was only after the appearance of Saito's related work [Sai26] that the key idea emerged. More specifically, the idea of deforming after the base change (1.11.1), which plays a central role in the present paper, was inspired by the construction of non-acyclicity classes in [YZ25, 4.2–4.3].

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Notation and Conventions.

- (1) All schemes are separated and of finite type over the fixed perfect field k of characteristic $p > 0$, unless otherwise stated.
- (2) The coefficient ring Λ is a finite local ring killed by a power of a prime $\ell \neq p$. The notation $D_{\text{ctf}}^b(X, \Lambda)$ denotes the derived (∞) -category of constructible complexes of finite Tor dimension.
- (3) We fix a nontrivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$. Let $\mathcal{L}_\psi$ be the Artin–Schreier sheaf on $\mathbb{A}_k^1$ corresponding to the character ψ . For a morphism $f : X \rightarrow \mathbb{A}_k^1$, we put $\mathcal{L}_\psi(f) = f^* \mathcal{L}_\psi$.
- (4) For a separated morphism $a : X \rightarrow S$, write $\mathcal{K}_{X/S} = Ra^! \Lambda$ and $\mathbb{D}_{X/S}(-) = R\mathcal{H}om(-, \mathcal{K}_{X/S})$. We write $\mathbb{D}_X$ when the base is $\text{Spec } k$.
- (5) For a smooth morphism $f : X \rightarrow Y$, the symbols $T(X/Y)$ and $T^*(X/Y)$ denote the relative tangent bundle and the relative cotangent bundle.
- (6) For a closed immersion $i : Z \hookrightarrow X$ between smooth schemes, the notation $T_Z X$ and $T_Z^* X$ denote the normal bundle and the conormal bundle.
- (7) Unless explicitly stated otherwise, we omit R and L from the notation for derived functors.

2. DEFORMATION TO THE NORMAL BUNDLE AND THE VERDIER SPECIALIZATION FUNCTOR

2.1. We first recall the construction of the deformation to the normal cone. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes defined by the quasi-coherent ideal sheaf $I \subseteq \mathcal{O}_X$, and let τ denote the coordinate on $\mathbb{A}^1$. Note that in this case i is a regular immersion.

The deformation of X to the normal bundle along $Z \hookrightarrow X$ is the scheme²

$$(2.1.1) \quad D_Z(X) = \operatorname{Spec}_X(\mathcal{R}_{I/\mathcal{O}_X}),$$

where $\mathcal{R}_{I/\mathcal{O}_X}$ is the extended Rees algebra

$$(2.1.2) \quad \mathcal{R}_{I/\mathcal{O}_X} = \mathcal{O}_X[\tau, I\tau^{-1}] = \bigoplus_{m \in \mathbb{Z}} I^m \tau^{-m} \subset \mathcal{O}_X[\tau, \tau^{-1}],$$

with the convention that $I^m = \mathcal{O}_X$ for $m \leq 0$. Equivalently, $D_Z(X)$ is the complement of the strict transform of $X \times \{0\}$ in the blow-up $\operatorname{Bl}_{Z \times \{0\}}(X \times \mathbb{A}^1)$.

2.2. The element τ defines a morphism

$$(2.2.1) \quad D_Z(X) \rightarrow \mathbb{A}^1,$$

which is smooth by [Sai26, Lemma 1.2.2]. The inverse image of $\mathbb{G}_m$ is canonically isomorphic to $X \times \mathbb{G}_m$, and the fiber over 0 is the normal bundle

$$(2.2.2) \quad T_Z X = \operatorname{Spec}_Z(\operatorname{Sym}_{\mathcal{O}_Z}(I/I^2)).$$

Together with the natural projection $D_Z(X) = \operatorname{Spec}_X(\mathcal{R}_{I/\mathcal{O}_X}) \rightarrow X$, the morphism (2.2.1) gives

$$(2.2.3) \quad \pi_{Z/X} : D_Z(X) \rightarrow X \times \mathbb{A}^1.$$

Lemma 2.3. *Let*

$$(2.3.1) \quad \begin{array}{ccc} Z & \xhookrightarrow{i} & X \\ \downarrow g & & \downarrow f \\ W & \xhookrightarrow{j} & Y \end{array}$$

be a commutative diagram of smooth schemes such that the horizontal arrows are closed immersions.

(1) *There is a natural morphism*

$$(2.3.2) \quad D_g(f) : D_Z(X) \rightarrow D_W(Y),$$

which fits into a commutative diagram

$$(2.3.3) \quad \begin{array}{ccc} D_Z(X) & \xrightarrow{D_g(f)} & D_W(Y) \\ \downarrow \pi_{Z/X} & & \downarrow \pi_{W/Y} \\ X \times \mathbb{A}^1 & \xrightarrow{f \times \operatorname{id}} & Y \times \mathbb{A}^1 \end{array}$$

(2) *If f and g are smooth, then $D_g(f)$ is smooth.*

(3) *If f and g are closed immersions and if (2.3.1) is cartesian, then $D_g(f)$ is a closed immersion.*

(4) *If (2.3.1) is cartesian and $\mathcal{O}_Z = \mathcal{O}_X \otimes_{\mathcal{O}_Y}^L \mathcal{O}_W$ (eg. f is flat), then (2.3.3) is cartesian.*

Proof. Consider the diagram

$$(2.3.4) \quad \begin{array}{ccc} Z \times \mathbb{A}^1 & \hookrightarrow & X \times \mathbb{A}^1 \\ \downarrow & & \downarrow \\ W \times \mathbb{A}^1 & \hookrightarrow & Y \times \mathbb{A}^1. \end{array}$$

The assertions follow by applying [Sai26, (1.50) and Lemma 1.2.3] to this diagram. $\square$

²In [Sai26], Saito denotes this $D_Z(X)$ by $D_{Z \times \mathbb{A}^1}(X \times \mathbb{A}^1)$ or $A_Z(X)$.

2.4. Consider the diagram (2.3.1). Its induced morphism $D_g(f)$ in (2.3.2) is defined over $\mathbb{A}^1$. Over $\mathbb{G}_m$, the morphism (2.3.2) is identified with $f \times \text{id} : X \times \mathbb{G}_m \rightarrow Y \times \mathbb{G}_m$:

$$(2.4.1) \quad \begin{array}{ccc} D_Z(X) \times_{\mathbb{A}^1} \mathbb{G}_m & \xrightarrow{D_g(f) \times \text{id}} & D_W(Y) \times_{\mathbb{A}^1} \mathbb{G}_m \\ \parallel & & \parallel \\ X \times \mathbb{G}_m & \xrightarrow{f \times \text{id}} & Y \times \mathbb{G}_m. \end{array}$$

On the special fiber, the morphism (2.3.2) is the natural map $T_Z X \rightarrow T_W Y$ between normal bundles:

$$(2.4.2) \quad \begin{array}{ccc} D_Z(X) \times_{\mathbb{A}^1} \{0\} & \xrightarrow{D_g(f) \times \text{id}} & D_W(Y) \times_{\mathbb{A}^1} \{0\} \\ \parallel & & \parallel \\ T_Z X & \xrightarrow{T_g f} & T_W Y. \end{array}$$

Remark 2.5. Let $i : Z \hookrightarrow X$ be a closed immersion between smooth schemes. The structure morphism $\pi_{Z/X} : D_Z(X) \rightarrow X \times \mathbb{A}^1$ can be described by the following diagram:

$$(2.5.1) \quad \begin{array}{ccc} D_Z(X) & \xrightarrow{\pi_{Z/X}} & X \times \mathbb{A}^1 \\ \parallel & & \downarrow \simeq \\ D_Z(X) & \xrightarrow{D_i(\text{id})} & D_X(X). \end{array}$$

For convenience, we recall the following result.

Lemma 2.6. *Let $f : Y \rightarrow X$ be a smooth morphism of relative dimension n , and let $i : X \hookrightarrow Y$ be a regular immersion such that $f \circ i = \text{id}$.*

For every geometric point x of X , there exist an open neighborhood U of x in X and an open neighborhood $V \subset f^{-1}(U)$ of $i(U)$ in Y , together with an étale morphism $V \rightarrow U \times \mathbb{A}^n$ through which $f|_V : V \rightarrow U$ factors, such that the following diagram

$$(2.6.1) \quad \begin{array}{ccc} U & \xrightarrow{i} & V \\ \parallel & & \downarrow \\ U & \xrightarrow{0} & U \times \mathbb{A}^n, \end{array}$$

is cartesian, where the lower horizontal morphism is the zero section.

Proof. The result follows by applying [Fu11, Theorem 2.5.8] to the diagram

$$(2.6.2) \quad \begin{array}{ccc} X & \xrightarrow{i} & Y \\ & \searrow & \downarrow f \\ & & X. \end{array}$$

□

Lemma 2.7. *Let $j : Z \hookrightarrow X$ and $i : X \hookrightarrow Y$ be closed immersions between smooth schemes, and let $f : Y \rightarrow X$ be a smooth morphism such that $f \circ i = \text{id}$. Then the following diagram is cartesian:*

$$(2.7.1) \quad \begin{array}{ccc} D_Z(Y) & \xrightarrow{D_{\text{id}}(f)} & D_Z(X) \\ \downarrow D_j(\text{id}) & & \downarrow \pi_{Z/X} \\ D_X(Y) & \xrightarrow{(f \times \text{id}) \circ \pi_{X/Y}} & X \times \mathbb{A}^1. \end{array}$$

Proof. Since the formation of deformation spaces is compatible with étale base change, we may work étale locally on X and Y .

Let τ be the coordinate on $\mathbb{A}^1$. Over $\tau \neq 0$, the diagram (2.7.1) is isomorphic to

$$(2.7.2) \quad \begin{array}{ccc} Y \times \mathbb{G}_m & \xrightarrow{f \times \text{id}} & X \times \mathbb{G}_m \\ \downarrow \text{id} & & \downarrow \text{id} \\ Y \times \mathbb{G}_m & \xrightarrow{f \times \text{id}} & X \times \mathbb{G}_m, \end{array}$$

which is cartesian.

It remains to work near the special fiber. The projection $T_X Y \rightarrow Y$ factors through $i : X \hookrightarrow Y$. So we only need to consider geometric points y of Y lying in $i(X)$.

Using the smoothness of f together with the section i , we may therefore reduce to the following case by Lemma 2.6

$$(2.7.3) \quad X = \text{Spec} A, \quad Y = X \times \mathbb{A}^n = \text{Spec} A[x_1, \dots, x_n],$$

where i is the zero section and f is the projection. Let $I \subset A$ be the defining ideal of $Z \subset X$. By (2.1.1), we have

$$(2.7.4) \quad D_Z(X) = \text{Spec} A[t, I/t].$$

The ideal defining $X \subseteq Y$ is $J = (x_1, \dots, x_n)$, and hence

$$(2.7.5) \quad D_X(Y) = \text{Spec} A[x_1, \dots, x_n, t, x_1/t, \dots, x_n/t].$$

Write $u_a = x_a/t$, so that $x_a = tu_a$. This gives a canonical isomorphism

$$(2.7.6) \quad D_X(Y) \simeq \text{Spec} A[t, u_1, \dots, u_n].$$

Notice that the ideal defining $Z \subset Y$ is $K = IA[x_1, \dots, x_n] + (x_1, \dots, x_n)$. Consequently, we have

$$(2.7.7) \quad \begin{aligned} D_Z(Y) &= \text{Spec} A[x_1, \dots, x_n, t, I/t, x_1/t, \dots, x_n/t] \\ &\simeq \text{Spec} A[t, I/t, u_1, \dots, u_n]. \end{aligned}$$

On the other hand, we have

$$(2.7.8) \quad \begin{aligned} D_Z(X) \times_{X \times \mathbb{A}^1} D_X(Y) &= \text{Spec} \left(A[t, I/t] \otimes_{A[t]} A[t, u_1, \dots, u_n] \right) \\ &\simeq \text{Spec} A[t, I/t, u_1, \dots, u_n] \simeq D_Z(Y). \end{aligned}$$

Hence the diagram (2.7.1) is cartesian. $\square$

Lemma 2.8. *Let $i : Z \hookrightarrow X$ and $j : W \hookrightarrow Y$ be closed immersions between smooth schemes over k . We have a natural isomorphism*

$$(2.8.1) \quad D_{Z \times W}(X \times Y) \simeq D_Z(X) \times_{\mathbb{A}^1} D_W(Y).$$

Proof. We work étale locally on X and on Y , and then we may assume that $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$ are affine. Let $I \subseteq A$ be the ideal defining Z in X , $J \subseteq B$ be the ideal defining W in Y , and

$$(2.8.2) \quad K = I \otimes_k B + A \otimes_k J,$$

be the ideal defining $Z \times W$ in $X \times Y$. It suffices to prove that

$$(2.8.3) \quad \mathcal{R}_{K/A \otimes_k B} \simeq \mathcal{R}_{I/A} \otimes_{k[\tau]} \mathcal{R}_{J/B}.$$

By definition, both sides of (2.8.3) are subrings of

$$(2.8.4) \quad A \otimes_k B[\tau, \tau^{-1}] \simeq A[\tau, \tau^{-1}] \otimes_{k[\tau]} B[\tau, \tau^{-1}]$$

generated by $A \otimes_k B[\tau]$, $(I \otimes_k B)\tau^{-1}$ and $(A \otimes_k J)\tau^{-1}$. $\square$

2.9. We recall the Verdier specialization functor from [Ver83]. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes over a field k . We consider the diagram

$$(2.9.1) \quad \begin{array}{ccccccc} T_Z X & \longrightarrow & D_Z(X) & \longleftarrow & X \times \mathbb{G}_m & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{A}^1 & \longleftarrow & \mathbb{G}_m, & & \end{array}$$

where both squares are cartesian. Let η_0 be the generic point of the henselization of $\mathbb{A}^1$ at 0 and $\bar{\eta}_0$ a geometric point above η_0 . Let $R\Psi : D_c^b(X \times \mathbb{G}_m) \rightarrow D_c^b(T_Z X \times_k \eta_0)$ be the nearby cycles functor with respect to $D_Z(X) \rightarrow \mathbb{A}^1$, where $D_c^b(T_Z X \times_k \eta_0, \Lambda)$ denotes the derived category of constructible sheaves on $T_Z X$ with continuous action of $\text{Gal}(\bar{\eta}_0/\eta_0)$ compatible with the action on $T_Z X$. The Verdier specialization functor

$$(2.9.2) \quad \nu_i = \nu_{Z/X} : D_c^b(X, \Lambda) \rightarrow D_c^b(T_Z X \times_k \eta_0, \Lambda), F \mapsto R\Psi(F \boxtimes \Lambda),$$

is defined by $\nu_{Z/X} = R\Psi \circ \text{pr}_1^*$. We suppress the factor η_0 from the notation and simply write $\nu_{Z/X}$ as a functor from $D_c^b(X, \Lambda)$ to $D_c^b(T_Z X, \Lambda)$.

3. RADON TRANSFORM AND FOURIER ORTHOGONALITY

3.1. Let V be a fixed vector space over k of dimension $n + 1$ with $n \geq 1$, and put $P = \mathbb{P}(V)$ and $P^\vee = \mathbb{P}(V^\vee)$. Consider the incidence correspondence

$$(3.1.1) \quad Q = \{(x, H) \in P \times P^\vee : x \in H\} \subseteq P \times P^\vee.$$

Let $p : Q \rightarrow P$ and $p^\vee : Q \rightarrow P^\vee$ be the projection maps. The Radon transform and the dual Radon transform are, respectively, the functors

$$(3.1.2) \quad \text{Rad} = Rp_* p^* [n-1] : D_c^b(P, \Lambda) \rightarrow D_c^b(P^\vee, \Lambda),$$

$$(3.1.3) \quad \text{Rad}^\vee = Rp_* p^{\vee*} [n-1] : D_c^b(P^\vee, \Lambda) \rightarrow D_c^b(P, \Lambda).$$

We define the anti-Radon transform

$$(3.1.4) \quad \overline{\text{Rad}} = \text{Rad}^\vee(n-1) : D_c^b(P^\vee, \Lambda) \rightarrow D_c^b(P, \Lambda).$$

By absolute purity, $p^! = p^*[2n-2](n-1)$ and $p^{\vee!} = p^{\vee*}[2n-2](n-1)$. Consequently, $\overline{\text{Rad}}$ is both left and right adjoint to Rad .

It is well known that Rad and $\overline{\text{Rad}}$ do not define an adjoint equivalence for $n \geq 2$ in general. Nevertheless, the following retraction statement holds.

Proposition 3.2. *Let U, C denote the unit and counit of the adjunction $\text{Rad} \vdash \overline{\text{Rad}}$, and let $\bar{U}, \bar{C}$ denote the unit and counit of the adjunction $\overline{\text{Rad}} \vdash \text{Rad}$. For every $F \in D_c^b(P, \Lambda)$, the following identity holds:*

$$(3.2.1) \quad \bar{C}_F \circ U_F = (-1)^{n-1} \text{id}_F.$$

In particular, F is a retract of $\overline{\text{Rad}}(\text{Rad}F)$.

Proof. Unwinding the definitions, the left-hand side of (3.2.1) is the composite

$$(3.2.2) \quad \begin{aligned} F &\rightarrow Rp_* p^* F \rightarrow Rp_* p^{\vee!} Rp_*^\vee p^* F \\ &\simeq Rp_* p^{\vee*} Rp_*^\vee p^* F[2n-2](n-1) \\ &\rightarrow Rp_* p^* F[2n-2](n-1) \simeq Rp_* p^! F \rightarrow F. \end{aligned}$$

We first evaluate the two arrows involving $p^{\vee!}$. By [ILO14, Exposé XVI, Définition 2.3.1 and Proposition 2.3.2], the composite

$$(3.2.3) \quad p^* F \rightarrow p^{\vee!} Rp_*^\vee p^* F \simeq p^{\vee*} Rp_*^\vee p^* F[2n-2](n-1) \rightarrow p^* F[2n-2](n-1)$$

is given by cup product with the Chern class $c_{n-1}(T(Q/P^\vee))$ of the relative tangent bundle $T(Q/P^\vee)$. By the projection formula, we get

$$(3.2.4) \quad \overline{C}_F U_F = p_* c_{n-1}(T(Q/P^\vee)) \cup \text{id}_F.$$

It remains to compute the cohomology class $p_* c_{n-1}(T(Q/P^\vee)) \in H^0(P, \Lambda)$. Let $x = [L] \in P$ be a geometric point, where $L \subset V$ is a line. We have $p^{-1}(x) = \mathbb{P}((V/L)^\vee) \simeq \mathbb{P}^{n-1}$, and there is an exact sequence

$$(3.2.5) \quad 0 \rightarrow T(Q/P^\vee)|_{p^{-1}(x)} \rightarrow \text{Hom}(L, V/L) \otimes \mathcal{O}_{p^{-1}(x)} \rightarrow L^\vee \otimes \mathcal{O}_{p^{-1}(x)}(1) \rightarrow 0,$$

and proper base change gives

$$(3.2.6) \quad \begin{aligned} p_* c_{n-1}(T(Q/P^\vee))_x &= \int_{p^{-1}(x)} c_{n-1}(T(Q/P^\vee)|_{p^{-1}(x)}) \\ &= (-1)^{n-1} \int_{p^{-1}(x)} c_1(\mathcal{O}_{p^{-1}(x)}(1))^{n-1} = (-1)^{n-1}. \end{aligned}$$

Consequently, $p_* c_{n-1}(T(Q/P^\vee)) = (-1)^{n-1}$ (see also [SGA5, Exposé VII, Lemme 8.4.1]). $\square$

3.3. Consider the following cartesian diagrams:

$$(3.3.1) \quad \begin{array}{ccc} Q \times_P Q & \xrightarrow{\text{pr}_{p,1}} & Q \\ \downarrow \text{pr}_{p,2} & & \downarrow p \\ Q & \xrightarrow{p} & P, \end{array} \quad \begin{array}{ccc} Q \times_P Q & \xleftarrow{i_p} & Q \times Q \\ \downarrow h_p & & \downarrow p \times p \\ P & \xleftarrow{\Delta_P} & P \times P. \end{array}$$

By Lemma 2.3, the following diagram is cartesian:

$$(3.3.2) \quad \begin{array}{ccc} D_{Q \times_P Q}(Q \times Q) & \xrightarrow{D_{h_p}(p \times p)} & D_P(P \times P) \\ \downarrow & & \downarrow \\ Q \times Q \times \mathbb{A}^1 & \xrightarrow{p \times p \times \text{id}} & P \times P \times \mathbb{A}^1. \end{array}$$

Over the special fiber, the morphism $D_{h_p}(p \times p)$ is the natural projection

$$(3.3.3) \quad \begin{array}{ccc} T_{Q \times_P Q}(Q \times Q) & \xrightarrow{T_{h_p}(p \times p)} & T_P(P \times P) \\ \parallel & & \parallel \\ Q \times_P Q \times_P TP & \xrightarrow{h_p \times \text{id}_{TP} = \text{pr}_{TP}} & TP \end{array}$$

Let X be a smooth scheme. For $F_1, F_2 \in D_{\text{ctf}}^b(X, \Lambda)$, we define

$$(3.3.4) \quad \mathcal{H}_X(F_1, F_2) = R\mathcal{H}om_{X \times X}(\text{pr}_1^* F_1, \text{pr}_2^! F_2) \in D_{\text{ctf}}^b(X \times X, \Lambda).$$

This is the internal Hom object for cohomological correspondences; see [LZ22]. If $F_1 = F_2 = F$, we write $\mathcal{H}_F = \mathcal{H}_X(F, F)$. Let $\Delta_X : X \rightarrow X \times X$ be the diagonal morphism. Its normal bundle is naturally identified with the tangent bundle TX of X . Using Verdier specialization functor (2.9.2), we set (cf. [Sai26, (2.1)])

$$(3.3.5) \quad \nu \mathcal{H}om_X(F_1, F_2) = \nu_{\Delta_X} \mathcal{H}_X(F_1, F_2) \in D_{\text{ctf}}^b(TX, \Lambda).$$

Lemma 3.4. *Retain the notation of Subsection 3.3. For $G_1, G_2 \in D_{\text{ctf}}^b(Q, \Lambda)$, there are natural isomorphisms*

$$(3.4.1) \quad R(p \times p)_* \mathcal{H}_Q(G_1, G_2) = \mathcal{H}_P(Rp_* G_1, Rp_* G_2),$$

$$(3.4.2) \quad R(h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_Q(G_1, G_2) = \nu_{\Delta_P} \mathcal{H}_P(Rp_* G_1, Rp_* G_2).$$

Proof. By [LZ22, Proposition 2.11] and the Künneth formula,

$$(3.4.3) \quad \begin{aligned} R(p \times p)_* \mathcal{H}_Q(G_1, G_2) &= R(p \times p)_*(\mathbb{D}_Q G_1 \boxtimes G_2) \\ &= Rp_* \mathbb{D}_Q G_1 \boxtimes Rp_* G_2 = \mathbb{D}_P(Rp_* G_1) \boxtimes Rp_* G_2 = \mathcal{H}_P(Rp_* G_1, Rp_* G_2). \end{aligned}$$

By (3.3.2), the morphism $\mathbb{D}_{Q \times_P Q}(Q \times Q) \rightarrow \mathbb{D}_P(P \times P)$ is proper over $\mathbb{A}^1$. By proper base change for nearby cycles [Sai26, (1.53)], we obtain

$$(3.4.4) \quad R(h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_Q(G_1, G_2) = \nu_{\Delta_P} R(p \times p)_* \mathcal{H}_Q(G_1, G_2) = \nu_{\Delta_P} \mathcal{H}_P(Rp_* G_1, Rp_* G_2).$$

□

Lemma 3.5. *Let $F \in D_{\text{ctf}}^b(P, \Lambda)$, and set $G = p^{\vee*} \text{Rad}(F)n-1 \in D_{\text{ctf}}^b(Q, \Lambda)$. Then $\nu \mathcal{H}om_P(F, F)$ is a retract of $R(h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_G$.*

Proof. Note that

$$(3.5.1) \quad Rp_* G = Rp_* p^{\vee*} \text{Rad}(F)n-1 = \overline{\text{Rad}} \text{Rad}(F).$$

By Proposition 3.2, F is a retract of $Rp_* G$. The retraction is defined by

$$(3.5.2) \quad F \xrightarrow{i=\text{U}_F} Rp_* G \xrightarrow{r=(-1)^{n-1} \overline{\text{C}}_F} F,$$

where $ri = \text{id}$. Let $\mathcal{C} = \text{CohCorr}_k$ be the symmetric monoidal category of cohomological correspondences (c.f. Appendix A). By [LZ22, Lemma 2.8], $\mathcal{H}_F = \mathcal{H}om_{\mathcal{C}}((P; F), (P; F))$ is the endomorphism object in the category $\mathcal{C}$ of cohomological correspondences. Therefore, $\mathcal{H}_F$ is a retract of $\mathcal{H}_{Rp_* G}$. Indeed, the retraction is given by

$$(3.5.3) \quad \begin{array}{ccccc} \mathcal{H}om_{\mathcal{C}}((P; F), (P; F)) & \longrightarrow & \mathcal{H}om_{\mathcal{C}}((P; Rp_* G), (P; Rp_* G)) & \longrightarrow & \mathcal{H}om_{\mathcal{C}}((P; F), (P; F)) \\ \downarrow (\text{id}, i) & & \parallel & & \uparrow (\text{id}, r) \\ \mathcal{H}om_{\mathcal{C}}((P; F), (P; Rp_* G)) & \xrightarrow{(r, \text{id})} & \mathcal{H}om_{\mathcal{C}}((P; Rp_* G), (P; Rp_* G)) & \xrightarrow{(i, \text{id})} & \mathcal{H}om_{\mathcal{C}}((P; F), (P; Rp_* G)), \end{array}$$

where

$$(3.5.4) \quad \begin{aligned} (\text{id}, r) \circ (i, \text{id}) \circ (r, \text{id}) \circ (\text{id}, i) &= (\text{id}, r) \circ (ri, \text{id}) \circ (\text{id}, i) \\ &= (\text{id}, r) \circ (\text{id}, \text{id}) \circ (\text{id}, i) = (\text{id}, r) \circ (\text{id}, i) = (\text{id}, r \circ i) = (\text{id}, \text{id}). \end{aligned}$$

Applying the functor ν_{Δ_P} , we get that $\nu \mathcal{H}om_P(F, F) = \nu_{\Delta_P} \mathcal{H}_F$ is a retract of

$$\nu_{\Delta_P} \mathcal{H}_{Rp_* G} \stackrel{(3.4.2)}{\simeq} R(h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_G.$$

This finishes the proof. □

3.6. For later convenience, we recall a result of Beilinson on the identification of the projection of singular supports with non-ULA locus. Recall the Legendre transform [Bei16]:

$$(3.6.1) \quad \mathbb{P}(T^*P) \simeq Q \simeq \mathbb{P}(T^*P^{\vee}).$$

Lemma 3.7 ([Bei16, Theorem 3.2]). *Let $F \in D_{\text{ctf}}^b(P, \Lambda)$, and let $\text{SS}(F) \subset T^*P$ be the singular support of F . Let $E_p(p^{\vee*} \text{Rad}(F))$ be the non-ULA locus of $p^{\vee*} \text{Rad}(F)$ with respect to $p : Q \rightarrow P$, i.e., $E_p(p^{\vee*} \text{Rad}(F))$ is the smallest closed subset of Q outside which p is universally locally acyclic relatively to $p^{\vee*} \text{Rad}(F)$. Under the Legendre transform $\mathbb{P}(T^*P) = Q$, we have*

$$(3.7.1) \quad \mathbb{P}(\text{SS}(F)) = E_p(p^{\vee*} \text{Rad}(F)).$$

3.8. We recall the Fourier–Deligne transform. Let $a : V \rightarrow S$ be a vector bundle of rank r , and let $a^\vee : V^\vee \rightarrow S$ be its dual. Denote the projection maps by

$$(3.8.1) \quad \mathrm{pr}_V : V \times_S V^\vee \rightarrow V, \quad \mathrm{pr}_{V^\vee} : V \times_S V^\vee \rightarrow V^\vee,$$

and the canonical pairing by

$$(3.8.2) \quad \langle \cdot, \cdot \rangle : V \times_S V^\vee \rightarrow \mathbb{A}_S^1, \langle v, \xi \rangle = \xi(v).$$

Fix a nontrivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$, and denote by $\mathcal{L}_\psi$ the Artin–Schreier sheaf on $\mathbb{A}^1$. The Fourier–Deligne transform is defined by

$$(3.8.3) \quad \mathcal{F}_{\psi, V/S} : D_{\mathrm{ctf}}^b(V, \Lambda) \rightarrow D_{\mathrm{ctf}}^b(V^\vee, \Lambda), K \mapsto R\mathrm{pr}_{V^\vee!}(\mathrm{pr}_V^* K \otimes \langle \cdot, \cdot \rangle^* \mathcal{L}_\psi).$$

The Fourier transform of a complex pulled back from the base is supported on the zero section.

Lemma 3.9. [Sai26, Lemma 1.1.2] *Let $M \in D_{\mathrm{ctf}}^b(S, \Lambda)$ and let $a : V \rightarrow S$ be the structure map. Then we have*

$$(3.9.1) \quad \mathrm{supp} \mathcal{F}_{\psi, V/S}(a^* M) \subset 0_{V^\vee},$$

where $0_{V^\vee} = S \hookrightarrow V^\vee$ is the zero section.

3.10. Let $a : V \rightarrow S, b : W \rightarrow S$ be vector bundles, and let $\beta : W \rightarrow V$ be a linear morphism of vector bundles over S . Let add_V be the addition map

$$(3.10.1) \quad \mathrm{add}_V : V \times_S V \rightarrow V, (v_1, v_2) \mapsto v_1 + v_2.$$

The morphism β induces an action μ of W on V :

$$(3.10.2) \quad \mu = \mathrm{add}_V \circ (\beta \times \mathrm{id}_V) : W \times_S V \rightarrow V, (w, v) \mapsto v + \beta(w).$$

Write $\mathrm{pr}_V = b \times \mathrm{id}_V : W \times_S V \rightarrow V$ for the projection. For $K \in D_{\mathrm{ctf}}^b(V, \Lambda)$, a W -invariant structure on K is an isomorphism

$$(3.10.3) \quad \mu^* K \simeq \mathrm{pr}_V^* K.$$

If K admits a W -invariant structure, then we say that K is W -invariant.

Remark 3.11. If the isomorphism (3.10.3) satisfies the usual cocycle condition, then K is W -equivariant. We do not need this stronger condition.

Proposition 3.12. *Retain the notation of Subsection 3.10, and let $K \in D_{\mathrm{ctf}}^b(V, \Lambda)$ be W -invariant. Consider the dual morphism*

$$(3.12.1) \quad \beta^\vee : V^\vee \rightarrow W^\vee,$$

and its kernel $\ker(\beta^\vee) \subset V^\vee$. Then we have

$$(3.12.2) \quad \mathrm{supp}(\mathcal{F}_{\psi, V/S} K) \subset \ker(\beta^\vee).$$

Proof. Let (s, ξ) be a geometric point of $V^\vee$, i.e., s is a geometric point of S and ξ is a geometric covector in $V_s^\vee$. Suppose that $(s, \xi) \notin \ker(\beta^\vee)$, equivalently, that $\beta_s^\vee(\xi) \neq 0$. Choose a line $L \subset W_s$ such that $\beta_s|_L$ is injective and $\xi|_{\beta_s(L)} \neq 0$. Choose a direct-sum decomposition

$$(3.12.3) \quad \beta_s(L) \oplus Z = V_s$$

Denote by

$$(3.12.4) \quad i_L : L \hookrightarrow W_s, \quad i_{\beta_s(L)} : \beta_s(L) \hookrightarrow V_s, \quad i_Z : Z \hookrightarrow V_s,$$

the inclusions, and by

$$(3.12.5) \quad p_{\beta_s(L)} : V_s \rightarrow \beta_s(L), \quad p_Z : V_s \rightarrow Z,$$

the projections. Under the decomposition (3.12.3), addition induces the isomorphism

$$(3.12.6) \quad \text{add}_{V_s}|_{\beta_s(L) \times Z} : \beta_s(L) \times Z \xrightarrow{i_{\beta_s(L)} \times i_Z} V_s \times V_s \xrightarrow{\text{add}_{V_s}} V_s.$$

Since K is W -invariant, there is an isomorphism

$$(3.12.7) \quad \mu_s^* K|_{V_s} \simeq \text{pr}_{V_s}^* K|_{V_s} \quad \text{in } D_{\text{ctf}}^b(W_s \times V_s, \Lambda).$$

The action map fits into the commutative diagram

$$(3.12.8) \quad \begin{array}{ccc} & \beta_s(L) \times Z & \xrightarrow[\text{=}]{} \text{add}_{V_s}|_{\beta_s(L) \times Z} V_s \\ & \downarrow \text{id}_{\beta_s(L)} \times i_Z & \\ L \times V_s & \xrightarrow[\text{=}]{} \beta_s(L) \times V_s & \\ \downarrow i_L \times \text{id}_{V_s} & & \downarrow \text{add}|_{\beta_s(L) \times V_s} \\ W_s \times V_s & \xrightarrow{\mu_s} V_s, & \leftarrow \text{id}_{V_s} \end{array}$$

while the projections fit into the commutative diagram

$$(3.12.9) \quad \begin{array}{ccc} & \beta_s(L) \times Z & \xrightarrow[\text{=}]{} \text{add}_{V_s}|_{\beta_s(L) \times Z} V_s \\ & \downarrow \text{id}_{\beta_s(L)} \times i_Z & \\ L \times V_s & \xrightarrow[\text{=}]{} \beta_s(L) \times V_s & \\ \downarrow i_L \times \text{id}_{V_s} & & \downarrow a_s|_{\beta_s(L)} \times \text{id}_{V_s} \\ W_s \times V_s & \xrightarrow{\text{pr}_{V_s}} V_s. & \leftarrow i_Z \circ p_Z \end{array}$$

Applying $*$ -pullback along the outer paths in (3.12.8) and (3.12.9) to the isomorphism (3.12.7), we obtain

$$(3.12.10) \quad K|_{V_s} \simeq p_Z^* i_Z^* K|_{V_s} = \Lambda \boxtimes i_Z^* K|_{V_s} \quad \text{in } D_c^b(V_s, \Lambda) = D_c^b(\beta_s(L) \times Z, \Lambda).$$

In the following, we calculate the stalk by using the Künneth formula

$$(3.12.11) \quad \begin{aligned} (\mathcal{F}_{\psi, V/S} K)_{(s, \xi)} &= R\Gamma_c(V_s, K_s \otimes \mathcal{L}_{\psi}(\xi)) \\ &\simeq R\Gamma_c(\beta_s(L) \times Z, \mathcal{L}_{\psi}(\xi|_{\beta_s(L)}) \boxtimes (i_Z^* K_s \otimes \mathcal{L}_{\psi}(\xi|_Z))) \\ &\simeq R\Gamma_c(\beta_s(L), \mathcal{L}_{\psi}(\xi|_{\beta_s(L)})) \otimes_{\Lambda}^L R\Gamma_c(Z, i_Z^* K_s \otimes \mathcal{L}_{\psi}(\xi|_Z)). \end{aligned}$$

Since $\xi|_{\beta_s(L)} \neq 0$, the associated Artin–Schreier sheaf has vanishing compactly supported cohomology:

$$(3.12.12) \quad R\Gamma_c(\beta_s(L), \mathcal{L}_{\psi}(\xi|_{\beta_s(L)})) = 0,$$

which completes the proof. $\square$

Remark 3.13. Take $W = V$ and $\beta = \text{id}_V$. Then Proposition 3.12 implies Lemma 3.9.

Lemma 3.14. Let $a : V \rightarrow S$ and $b : W \rightarrow T$ be vector bundles. For $F \in D_{\text{ctf}}^b(V, \Lambda)$ and $G \in D_{\text{ctf}}^b(W, \Lambda)$, there is a natural isomorphism

$$(3.14.1) \quad \mathcal{F}_{\psi, V \times W/S \times T}(F \boxtimes G) = \mathcal{F}_{\psi, V/S}(F) \boxtimes \mathcal{F}_{\psi, W/T}(G).$$

Proof. We have natural isomorphisms

$$(3.14.2) \quad (V \times W) \times_{S \times T} (V \times W)^\vee = (V \times W) \times_{S \times T} (V^\vee \times W^\vee) = (V \times_S V^\vee) \times (W \times_T W^\vee).$$

Consider the following commutative diagram

$$(3.14.3) \quad \begin{array}{ccc} (V \times W) \times_{S \times T} (V \times W)^\vee & \xlongequal{\quad} & (V \times_S V^\vee) \times (W \times_T W^\vee) \\ \downarrow & & \downarrow \\ \mathbb{A}^1 & \xleftarrow{\quad \text{add} \quad} & \mathbb{A}^1 \times \mathbb{A}^1. \end{array}$$

Since $\text{add}^* \mathcal{L}_\psi = \mathcal{L}_\psi \boxtimes \mathcal{L}_\psi$, we obtain an isomorphism

$$(3.14.4) \quad \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{V \times W}) = \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_V) \boxtimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_W).$$

Now (3.14.1) is the following composition of isomorphisms

$$(3.14.5) \quad \begin{aligned} \mathcal{F}_{\psi, V \times W/S \times T}(F \boxtimes G) &= R(\text{pr}_{V^\vee \times W^\vee})!(\text{pr}_{V \times W}^*(F \boxtimes G) \otimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{V \times W})) \\ &= R(\text{pr}_{V^\vee} \times \text{pr}_{W^\vee})!((\text{pr}_V^* F \boxtimes \text{pr}_W^* G) \otimes (\mathcal{L}_\psi(\langle \cdot, \cdot \rangle_V) \boxtimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_W))) \\ &\stackrel{(1)}{=} R\text{pr}_{V^\vee}!(\text{pr}_V^* F \otimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_V)) \boxtimes R\text{pr}_{W^\vee}!(\text{pr}_W^* G \otimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_W)) \\ &= \mathcal{F}_{\psi, V/S}(F) \boxtimes \mathcal{F}_{\psi, W/T}(G), \end{aligned}$$

where (1) is an isomorphism by the Künneth formula. $\square$

Lemma 3.15. *Let $V \rightarrow S$ and $W \rightarrow S$ be vector bundles. Let $q : V \rightarrow W$ be a linear morphism of vector bundles. We have an action*

$$(3.15.1) \quad \mu = \text{add}_W \circ (q \times \text{id}) : V \times_S W \rightarrow W, (v, w) \mapsto q(v) + w.$$

Consider the diagrams

$$(3.15.2) \quad \begin{array}{ccc} V \times_S W & \xrightarrow{\mu} & W \\ \text{pr}_1 \swarrow & \downarrow \text{pr}_2 & \\ V & & W, \end{array} \quad \begin{array}{ccc} W \times_S W^\vee & \xrightarrow{\langle \cdot, \cdot \rangle_W} & \mathbb{A}^1 \\ \text{pr}_W \swarrow & \downarrow \text{pr}_{W^\vee} & \\ W & & W^\vee. \end{array}$$

For $F \in D_{\text{ctf}}^b(V, \Lambda)$ and $G \in D_{\text{ctf}}^b(W, \Lambda)$, we have a natural isomorphism

$$(3.15.3) \quad \mathcal{F}_{\psi, W/S}(R\mu_!(\text{pr}_1^* F \otimes \text{pr}_2^* G)) \simeq q^{\vee*} \mathcal{F}_{\psi, V/S}(F) \otimes \mathcal{F}_{\psi, W/S}(G).$$

Proof. This is a standard calculation for composition of Fourier-Mukai type transformations. We include a proof for completeness. We denote by

$$(3.15.4) \quad p_1 : V \times_S W \times_S W^\vee \rightarrow V, \quad p_2 : V \times_S W \times_S W^\vee \rightarrow W, \quad p_3 : V \times_S W \times_S W^\vee \rightarrow W^\vee$$

the projection morphisms. Consider the pullback diagram

$$(3.15.5) \quad \begin{array}{ccc} V \times_S W \times_S W^\vee & \xrightarrow{\mu \times \text{id}_{W^\vee}} & W \times_S W^\vee \\ \downarrow \text{pr}_{1,2} & & \downarrow \text{pr}_W \\ V \times_S W & \xrightarrow{\mu} & W. \end{array}$$

We calculate the left-hand side of (3.15.3) as follows:

$$(3.15.6) \quad \begin{aligned} &\mathcal{F}_{\psi, W/S}(R\mu_!(\text{pr}_1^* F \otimes \text{pr}_2^* G)) \\ &= R\text{pr}_{W^\vee}!(\text{pr}_W^* R\mu_!(\text{pr}_1^* F \otimes \text{pr}_2^* G) \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W) \\ &= R\text{pr}_{W^\vee}!(R(\mu \times \text{id}_{W^\vee})!(\text{pr}_{1,2}^*(\text{pr}_1^* F \otimes \text{pr}_2^* G) \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W)) \\ &= R\text{pr}_{W^\vee}!R(\mu \times \text{id}_{W^\vee})!(\text{pr}_{1,2}^*(\text{pr}_1^* F \otimes \text{pr}_2^* G) \otimes (\mu \times \text{id}_{W^\vee})^* \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W) \\ &= Rp_{3!}(p_1^* F \otimes p_2^* G \otimes \mathcal{L}_\psi(\langle \mu(\cdot, \cdot), \cdot \rangle_W)). \end{aligned}$$

We have commutative diagrams

$$(3.15.7) \quad \begin{array}{ccc} V \times_S W \times_S W^\vee & \xrightarrow{\mu \times \text{id}_{W^\vee}} & W \times_S W^\vee \xrightarrow{\langle \cdot, \cdot \rangle_W} \mathbb{A}^1 \\ \downarrow (\text{pr}_{1,3}, \text{pr}_{2,3}) & & \uparrow \text{add} \\ (V \times_S W^\vee) \times (W \times_S W^\vee) & \xrightarrow{\langle q(\cdot), \cdot \rangle_W \times \langle \cdot, \cdot \rangle_W} & \mathbb{A}^1 \times \mathbb{A}^1, \end{array}$$

$$(3.15.8) \quad \begin{array}{ccc} V \times_S W^\vee & \xrightarrow{\text{id}_V \times q^\vee} & V \times_S V^\vee \\ \downarrow q \times \text{id}_{W^\vee} & & \downarrow \langle \cdot, \cdot \rangle_V \\ W \times_S W^\vee & \xrightarrow{\langle \cdot, \cdot \rangle_W} & \mathbb{A}^1, \end{array}$$

and pullback diagrams

$$(3.15.9) \quad \begin{array}{ccc} V \times_S W^\vee & \xrightarrow{\text{id}_V \times q^\vee} & V \times_S V^\vee \\ \downarrow \text{pr}_2 & & \downarrow \text{pr}_{V^\vee} \\ W^\vee & \xrightarrow{q^\vee} & V^\vee, \end{array}$$

$$(3.15.10) \quad \begin{array}{ccc} V \times_S W \times_S W^\vee & \xrightarrow{\text{pr}_{1,3}} & V \times_S W^\vee \\ \downarrow \text{pr}_{2,3} & & \downarrow \text{pr}_2 \\ W \times_S W^\vee & \xrightarrow{\text{pr}_{W^\vee}} & W^\vee. \end{array}$$

Then we can calculate the right-hand side of (3.15.3) as follows:

$$(3.15.11) \quad \begin{aligned} & q^{\vee*} \mathcal{F}_{\psi, V/S}(F) \otimes \mathcal{F}_{\psi, W/S}(G) \\ &= q^{\vee*} R\text{pr}_{V^\vee!}(\text{pr}_V^* F \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_V) \otimes R\text{pr}_{W^\vee!}(\text{pr}_W^* G \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W) \\ &= R\text{pr}_{2!}((\text{id}_V \times q^\vee)^*(\text{pr}_V^* F \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_V) \otimes R\text{pr}_{W^\vee!}(\text{pr}_W^* G \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W)) \\ &= R\text{pr}_{2!}((\text{id}_V \times q^\vee)^*(\text{pr}_V^* F \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_V) \otimes \text{pr}_2^* R\text{pr}_{W^\vee!}(\text{pr}_W^* G \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W)) \\ &= R\text{pr}_{2!}((\text{id}_V \times q^\vee)^*(\text{pr}_V^* F \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_V) \otimes R(\text{pr}_{1,3}!) \text{pr}_{2,3}^*(\text{pr}_W^* G \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W)) \\ &= R\text{pr}_{2!} R(\text{pr}_{1,3}!) (\text{pr}_{1,3}^*(\text{id}_V \times q^\vee)^*(\text{pr}_V^* F \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_V) \otimes \text{pr}_{2,3}^*(\text{pr}_W^* G \otimes \mathcal{L}_\psi \langle \cdot, \cdot \rangle_W)) \\ &= R p_3! (p_1^* F \otimes p_2^* G \otimes \mathcal{L}_\psi (\langle \cdot, q^\vee(\cdot) \rangle_V \circ \text{pr}_{1,3} + \langle \cdot, \cdot \rangle_W \circ \text{pr}_{2,3})) \\ &= R p_3! (p_1^* F \otimes p_2^* G \otimes \mathcal{L}_\psi (\langle \mu(\cdot, \cdot), \cdot \rangle_W)). \end{aligned}$$

By (3.15.6) and (3.15.11), we get the isomorphism (3.15.3). $\square$

3.16. Now we construct an invariant structure on Verdier specializations. Let $f : X \rightarrow Y$ be a smooth morphism, and let $i : Z \hookrightarrow X$ be a regular immersion between smooth k -schemes. Let $\Delta_f : X \hookrightarrow X \times_Y X$ be the diagonal morphism. Applying Lemma 2.7 to the following diagram

$$(3.16.1) \quad \begin{array}{ccccc} Z & \xrightarrow{i} & X & \xrightarrow{\Delta_f} & X \times_Y X & \xrightarrow{\text{pr}_{f,2}} & X \\ & & & \searrow & \downarrow \text{pr}_{f,1} & & \downarrow f \\ & & & & X & \xrightarrow{f} & Y. \end{array}$$

we obtain a cartesian diagram

$$(3.16.2) \quad \begin{array}{ccc} D_Z(X \times_Y X) & \xrightarrow{D_{\text{id}}(\text{pr}_{f,1})} & D_Z(X) \\ \downarrow D_i(\text{id}) & & \downarrow \pi_{Z/X} \\ D_X(X \times_Y X) & \xrightarrow{\pi_{\Delta_f}} & X \times \mathbb{A}^1, \end{array}$$

where $\pi_{\Delta_f} = (\text{pr}_{f,1} \times \text{id}) \circ \pi_{X/X \times_Y X}$. The second projection $\text{pr}_{f,2} : X \times_Y X \rightarrow X$ induces the following action:

$$(3.16.3) \quad \tilde{\mu}_{Z/X/Y} : D_X(X \times_Y X) \times_{X \times \mathbb{A}^1} D_Z(X) \xleftarrow[\text{=}]{{(D_i(\text{id}), D_{\text{id}}(\text{pr}_{f,1}))}} D_Z(X \times_Y X) \xrightarrow{D_{\text{id}}(\text{pr}_{f,2})} D_Z(X).$$

Unwinding the definitions, on the generic fiber, (3.16.3) is given by

$$(3.16.4) \quad \text{pr}_{f,2} : X \times_Y X \rightarrow X,$$

whereas on the special fiber, (3.16.3) induces an action

$$(3.16.5) \quad \mu_{Z/X/Y} : T(X/Y)|_Z \times_Z T_Z X = T_{X/Y} \times_X T_Z X \rightarrow T_Z X.$$

Similarly, the morphism induced by $\text{pr}_{f,1}$ is the projection onto the $D_Z(X)$ -factor:

$$(3.16.6) \quad \text{pr}_{D_Z(X)} : D_X(X \times_Y X) \times_{X \times \mathbb{A}^1} D_Z(X) \xleftarrow[\text{=}]{{(D_i(\text{id}), D_{\text{id}}(\text{pr}_{f,1}))}} D_Z(X \times_Y X) \xrightarrow{D_{\text{id}}(\text{pr}_{f,1})} D_Z(X).$$

On the generic fiber, the morphism (3.16.6) is given by

$$(3.16.7) \quad \text{pr}_{f,1} : X \times_Y X \rightarrow X.$$

On the special fiber, (3.16.6) is the projection

$$(3.16.8) \quad \text{pr}_{T_Z X} : T(X/Y)|_Z \times_Z T_Z X = T(X/Y) \times_X T_Z X \rightarrow T_Z X,$$

where $T(X/Y)$ is the relative tangent bundle of $X \rightarrow Y$ and it is isomorphic to the normal bundle of $\Delta_f : X \rightarrow X \times_Y X$.

Lemma 3.17. *Retain the notation of Subsection 3.16. For $K \in D_{\text{ctf}}^b(Y, \Lambda)$, the complex $\nu_{Z/X}(f^* K) \in D_{\text{ctf}}^b(T_Z X, \Lambda)$ is $T(X/Y)|_Z$ -invariant for the action of $T(X/Y)|_Z$ on $T_Z X$ defined by (3.16.5).*

Proof. Since $f \circ \text{pr}_{f,1} = f \circ \text{pr}_{f,2}$, there is an isomorphism

$$(3.17.1) \quad \text{pr}_{f,1}^* f^* K \simeq \text{pr}_{f,2}^* f^* K.$$

By Lemma 2.3, (3.16.3) and (3.16.6) are smooth morphisms over $\mathbb{A}^1$. Applying [Sai26, (1.52)] to these two morphisms yields natural isomorphisms

$$(3.17.2) \quad \nu_{Z/X \times_Y X} \text{pr}_{f,2}^* f^* K \simeq \mu_{Z/X/Y}^* \nu_{Z/X} f^* K,$$

$$(3.17.3) \quad \nu_{Z/X \times_Y X} \text{pr}_{f,1}^* f^* K \simeq \text{pr}_{T_Z X}^* \nu_{Z/X} f^* K.$$

Combining (3.17.1), (3.17.2), and (3.17.3), we obtain an isomorphism

$$(3.17.4) \quad \mu_{Z/X/Y}^* \nu_{Z/X} f^* K \simeq \text{pr}_{T_Z X}^* \nu_{Z/X} f^* K \quad \text{in } D_{\text{ctf}}^b(T(X/Y) \times_X T_Z X, \Lambda).$$

This proves the assertion. $\square$

3.18. Retain the notation of Subsection 3.3. We apply Lemma 3.17 to the following diagram

$$(3.18.1) \quad \begin{array}{ccccccc} Q \times_P Q & \xrightarrow{i_P} & Q \times Q & \xrightarrow{\Delta} & (Q \times Q) \times_{P^\vee \times P^\vee} (Q \times Q) & \xrightarrow{\text{pr}_2} & Q \times Q \\ & & & \searrow & \downarrow \text{pr}_1 & & \downarrow p^\vee \times p^\vee \\ & & & & Q \times Q & \xrightarrow{p^\vee \times p^\vee} & P^\vee \times P^\vee. \end{array}$$

We now describe the action (3.16.5) in this setting. There are canonical identifications

$$(3.18.2) \quad T(Q \times Q/P^\vee \times P^\vee) = \text{pr}_{Q,1}^* T(Q/P^\vee) \oplus \text{pr}_{Q,2}^* T(Q/P^\vee) \quad \text{over } Q \times Q,$$

$$(3.18.3) \quad T_{Q \times_P Q}(Q \times Q) = Q \times_P Q \times_P TP \quad \text{over } Q \times_P Q.$$

Define a vector bundle W over $Q \times_P Q$ by

$$(3.18.4) \quad W = T(Q \times Q/P^\vee \times P^\vee)|_{Q \times_P Q} = \text{pr}_{p,1}^* T(Q/P^\vee) \oplus \text{pr}_{p,2}^* T(Q/P^\vee).$$

Under these identifications, the action (3.16.5) becomes an action of W on $Q \times_P Q \times_P TP$ over $Q \times_P Q$:

$$(3.18.5) \quad \mu : W \times_P TP = W \times_{Q \times_P Q} (Q \times_P Q \times_P TP) \rightarrow Q \times_P Q \times_P TP.$$

More precisely, the action (3.18.5) is induced by the linear morphism

$$(3.18.6) \quad \beta : W \rightarrow Q \times_P Q \times_P TP, (w_1, w_2) \mapsto Tp(w_2) - Tp(w_1),$$

namely, the difference of the two morphisms

$$(3.18.7) \quad Tp \circ \text{pr}_2 : W \rightarrow \text{pr}_2^* T(Q/P^\vee) \rightarrow T(Q/P^\vee) \xrightarrow{Tp} TP,$$

$$(3.18.8) \quad Tp \circ \text{pr}_1 : W \rightarrow \text{pr}_1^* T(Q/P^\vee) \rightarrow T(Q/P^\vee) \xrightarrow{Tp} TP.$$

Here $Tp : T(Q/P^\vee) \rightarrow TP$ is the composition $T(Q/P^\vee) \rightarrow T(P \times P^\vee/P^\vee) \simeq TP \times P^\vee \xrightarrow{\text{pr}} TP$.

Lemma 3.19. *Retain the notation of Subsection 3.18.*

(1) *Let $F \in D_{\text{ctf}}^b(P, \Lambda)$ and put $G = p^{\vee*} \text{Rad}(F)n-1 \in D_{\text{ctf}}^b(Q, \Lambda)$. Then $\nu_{i_P}(\mathcal{H}_G)$ on $Q \times_P Q \times_P TP$ is W -invariant.*

(2) *Consider the dual morphism*

$$(3.19.1) \quad \beta^\vee : Q \times_P Q \times_P T^*P \rightarrow W^\vee = \text{pr}_{p,1}^* T^*(Q/P^\vee) \oplus \text{pr}_{p,2}^* T^*(Q/P^\vee).$$

*Let (x, y_1, y_2, ξ) be a geometric point of $Q \times_P Q \times_P T^*P$, with*

$$(3.19.2) \quad x \in P, \quad y_1 = [H_1] \in Q_x, \quad y_2 = [H_2] \in Q_x, \quad H_1, H_2 \subset P, \quad \text{and } \xi \in T_x^*P.$$

Suppose that $\xi \neq 0$. Then $(x, y_1, y_2, \xi) \in \ker(\beta^\vee)$ if and only if

$$(3.19.3) \quad y_1 = y_2, \quad \text{and } T_x H_1 = T_x H_2 = \ker(\xi).$$

Proof. We have isomorphisms

$$(3.19.4) \quad \begin{aligned} \mathbb{D}_Q G &= \mathbb{D}_Q((p^\vee)^* \text{Rad}(F)n-1) = (p^\vee)^\dagger \mathbb{D}_{P^\vee} \text{Rad}(F)1-n \\ &= (p^\vee)^* \mathbb{D}_{P^\vee} \text{Rad}(F)[n-1], \end{aligned}$$

$$(3.19.5) \quad \mathcal{H}_G = \mathbb{D}_Q G \boxtimes G = (p^\vee \times p^\vee)^*(\mathbb{D}_{P^\vee} \text{Rad}(F) \boxtimes \text{Rad}(F))[2n-2](n-1).$$

Now (1) follows from Lemma 3.17. We prove (2). Consider the morphism

$$(3.19.6) \quad \beta_{(y_1, y_2)} : W_{(y_1, y_2)} = T_{y_1}(Q/P^\vee) \oplus T_{y_2}(Q/P^\vee) \rightarrow T_x P.$$

For every $y = [H] \in Q_x$, the morphism $Tp : T_y(Q/P^\vee) \rightarrow T_x P$ induces an isomorphism

$$(3.19.7) \quad Tp : T_y(Q/P^\vee) = T_x H.$$

The image of this morphism is therefore

$$(3.19.8) \quad \text{im}\beta_{(y_1, y_2)} = T_x H_1 + T_x H_2 \subset T_x P.$$

The following conditions are equivalent for nonzero ξ :

$$(3.19.9) \quad \xi \in \ker(\beta^\vee) \iff \xi(\text{im}\beta_{(y_1, y_2)}) = 0$$

$$(3.19.10) \quad \iff \xi(T_x H_1) = 0 \quad \text{and} \quad \xi(T_x H_2) = 0$$

$$(3.19.11) \quad \iff y_1 = y_2 \quad \text{and} \quad T_x H_1 = T_x H_2 = \ker(\xi).$$

This completes the proof. $\square$

4. COMPARISON ON SINGULAR SUPPORTS

4.1. Let X be a smooth scheme over k . The dual of the tangent bundle $TX \rightarrow X$ is the cotangent bundle $T^*X \rightarrow X$. For $F_1, F_2 \in D_{\text{ctf}}^b(X, \Lambda)$, define the microlocal Hom complex by

$$(4.1.1) \quad \mu\text{Hom}_X(F_1, F_2) = \mathcal{F}_{\psi, TX/X}(\nu\text{Hom}_X(F_1, F_2)) \quad \text{in} \quad D_{\text{ctf}}^b(T^*X, \Lambda).$$

The microlocal singular support of $F \in D_{\text{ctf}}^b(X, \Lambda)$ is

$$(4.1.2) \quad \text{SS}_\mu(F) = \text{supp}\mu\text{Hom}_X(F, F) \subseteq T^*X.$$

Theorem 4.2. *Let Λ be a finite local ring killed by a power of a prime $\ell \neq p$. Let X be a smooth k -scheme, and let $F \in D_{\text{ctf}}^b(X, \Lambda)$. Then*

$$(4.2.1) \quad \text{SS}_\mu(F) \subseteq \text{SS}(F).$$

The proof of Theorem 4.2 is given in Subsection 4.9.

4.3. We first reduce the assertion to the case of projective spaces.

Definition 4.4. (1) Let X be a smooth k -scheme. We say that a complex $F \in D_{\text{ctf}}^b(X, \Lambda)$ satisfies the singular support inclusion if

$$(4.4.1) \quad \text{SS}_\mu(F) \subset \text{SS}(F).$$

(2) Let P be a projective space over k , and retain the notation of Subsection 3.1. Set

$$(4.4.2) \quad \rho : (T^*P)^\times = T^*P \setminus T_P^*P \rightarrow \mathbb{P}(T^*P) \simeq Q,$$

which is the composite of the projectivization map and the Legendre transform. We say that a complex $F \in D_{\text{ctf}}^b(P, \Lambda)$ satisfies the projective singular support inclusion if

$$(4.4.3) \quad \text{SS}_\mu(F)^\times = \text{SS}_\mu(F) \setminus T_P^*P \subset \rho^{-1}(E_p(p^{\vee*}\text{Rad}(F))) \stackrel{(3.7.1)}{=} \rho^{-1}(\mathbb{P}(\text{SS}(F))).$$

Lemma 4.5. *Assume that, for every $n \geq 1$, every $(n+1)$ -dimensional vector space V , and every $F \in D_{\text{ctf}}^b(\mathbb{P}(V), \Lambda)$, the complex F satisfies the projective singular support inclusion.*

Then, for every smooth scheme X over k and every $F \in D_{\text{ctf}}^b(X, \Lambda)$, the complex F satisfies the singular support inclusion. Equivalently, Theorem 4.2 holds.

Proof. When $\dim X \leq 1$, the Theorem 4.2 follows from [Sai26, Lemma 2.1.2.(3)].

Combining [Sai26, Lemma 2.3.1 and 2.4.1] with [Bei16, Lemma 2.3 and 2.5], we may assume that $X = P = \mathbb{P}(V)$ is a projective space. Let $F \in D_{\text{ctf}}^b(P, \Lambda)$. Since F satisfies the projective singular support inclusion, and both $\text{SS}_\mu(F)$ and $\text{SS}(F)$ are conical (cf. [Sai26, Lemma 2.1.2] and [Bei16, 1.3]), we have

$$(4.5.1) \quad \text{SS}_\mu(F)^\times \subset \text{SS}(F)^\times.$$

By [Bei16, Lemma 2.3], the base of $\mathrm{SS}(F)$ is $\mathrm{supp}(F)$. If F vanishes on an open neighborhood U of $x \in P$, then $\mathcal{H}_F$ vanishes on $U \times U$. Consequently, $\nu\mathcal{H}om_X(F, F)$ vanishes on TU and $\mu\mathcal{H}om_X(F, F)$ vanishes on T^*U . It follows that

$$(4.5.2) \quad \mathrm{SS}_\mu(F) \cap T_P^*P \subset \mathrm{supp}(F) = \mathrm{SS}(F) \cap T_P^*P.$$

Now the result follows from (4.5.1) and (4.5.2). $\square$

4.6. For the remainder of this section, let P be a projective space over k and let $F \in D_{\mathrm{ctf}}^b(P, \Lambda)$. Retain the notation of Subsection 3.1, and set

$$(4.6.1) \quad G = p^{\vee*}\mathrm{Rad}(F)n-1 \in D_{\mathrm{ctf}}^b(Q, \Lambda).$$

Universal local acyclicity is unaffected by cohomological shifts and Tate twists. Therefore,

$$(4.6.2) \quad E_p(p^{\vee*}\mathrm{Rad}(F)) = E_p(G)$$

is the non-ULA locus of G with respect to $p : Q \rightarrow P$.

Lemma 4.7. *Recall the notation $i_p : Q \times_P Q \hookrightarrow Q \times Q$. Let $U \subset Q$ be an open subscheme. If $p|_U : U \rightarrow P$ is universally locally acyclic relatively to $G|_U$, then the natural morphism*

$$(4.7.1) \quad i_p^*\mathcal{H}_G \boxtimes \Lambda_{TP} \rightarrow \nu_{i_p}\mathcal{H}_G \quad \text{in } D_{\mathrm{ctf}}^b(Q \times_P Q \times_P TP, \Lambda)$$

restricts to an isomorphism on $U \times_P U \times_P TP$.

Proof. By [LZ22, Theorem 2.16], the morphism $p|_U \times p|_U : U \times U \rightarrow P \times P$ is universally locally acyclic relatively to the complex

$$(4.7.2) \quad \mathcal{H}_G|_{U \times U} = \mathbb{D}_U(G|_U) \boxtimes (G|_U) \in D_{\mathrm{ctf}}^b(U \times U, \Lambda).$$

Consider the diagram (3.3.1). By Lemma 2.3, the following squares are cartesian:

$$(4.7.3) \quad \begin{array}{ccccc} D_{U \times_P U}(U \times U) & \hookrightarrow & D_{Q \times_P Q}(Q \times Q) & \xrightarrow{D_{h_p}(p \times p)} & D_P(P \times P) \\ \downarrow & & \downarrow & & \downarrow \\ U \times U & \hookrightarrow & Q \times Q & \xrightarrow{p \times p} & P \times P. \end{array}$$

Then $D_{h_p}(p \times p)|_{D_{U \times_P U}(U \times U)}$ is universally locally acyclic relatively to $\mathcal{H}_G|_{D_{U \times_P U}(U \times U)}$. Since $D_P(P \times P) \rightarrow \mathbb{A}^1$ is smooth, it follows from [SGA4 $\frac{1}{2}$, Th. finitude, Lemme 2.14] that $D_{U \times_P U}(U \times U) \rightarrow \mathbb{A}^1$ is universally locally acyclic relatively to $\mathcal{H}_G|_{D_{U \times_P U}(U \times U)}$. Taking nearby cycles yields natural isomorphisms

$$(4.7.4) \quad (i_p^*\mathcal{H}_G)|_{U \times_P U} \boxtimes \Lambda_{TP} = \mathcal{H}_G|_{N_{U \times_P U/U \times U}} \xrightarrow{\cong} \nu_{U \times_P U/U \times U}\mathcal{H}_G = (\nu_{i_p}\mathcal{H}_G)|_{U \times_P U \times_P TP},$$

which is the restriction of (4.7.1) to $U \times_P U \times_P TP$. $\square$

Recall the notation $h_p : Q \times_P Q \rightarrow P$ in (3.3.1) and that $h_p \times \mathrm{id}_{TP}$ in (3.3.3) denotes the projection

$$(4.7.5) \quad h_p \times \mathrm{id}_{TP} = \mathrm{pr}_{TP} : Q \times_P Q \times_P TP \rightarrow TP.$$

Lemma 4.8. *Let $U \subset Q$ be an open subscheme. Suppose that $p|_U$ is universally locally acyclic relatively to $G|_U$. Then the complex*

$$(4.8.1) \quad \mathcal{F}_{\psi, TP/P}(R(h_p \times \mathrm{id}_{TP})_* \nu_{i_p}\mathcal{H}_G)$$

vanishes on $\rho^{-1}(U)$.

Proof. Let Z be the closed complement of $U \times_P U$ in $Q \times_P Q$. Write

$$(4.8.2) \quad U \times_P U \xrightarrow{J} Q \times_P Q \xleftarrow{I} Z,$$

and denote by

$$(4.8.3) \quad J_{TP} = J \times \text{id}_{TP} : U \times_P U \times_P TP \hookrightarrow Q \times_P Q \times_P TP,$$

$$(4.8.4) \quad I_{TP} = I \times \text{id}_{TP} : Z \times_P TP \hookrightarrow Q \times_P Q \times_P TP$$

their base changes. There is a localization triangle

$$(4.8.5) \quad J_{TP!} J_{TP}^* \nu_{i_p} \mathcal{H}_G \rightarrow \nu_{i_p} \mathcal{H}_G \rightarrow I_{TP*} I_{TP}^* \nu_{i_p} \mathcal{H}_G \xrightarrow{+1}.$$

We first consider the open term. By Lemma 4.7, the complex

$$(4.8.6) \quad J_{TP}^* \nu_{i_p} \mathcal{H}_G = (i_p^* \mathcal{H}_G)|_{U \times_P U} \boxtimes \Lambda_{TP}$$

is relatively constant along the bundle $U \times_P U \times_P TP \rightarrow U \times_P U$. Hence Lemma 3.9 gives

$$(4.8.7) \quad \text{supp } \mathcal{F}_{\psi, U \times_P U \times_P TP / U \times_P U} (J_{TP}^* \nu_{i_p} \mathcal{H}_G) \subset U \times_P U \times_P 0_{T^*P}.$$

The following diagrams are cartesian:

$$(4.8.8) \quad \begin{array}{ccc} U \times_P U \times_P TP & \longrightarrow & U \times_P U \\ \downarrow h_p J \times \text{id}_{TP} & & \downarrow h_p J \\ TP & \longrightarrow & P, \end{array} \quad \begin{array}{ccc} U \times_P U \times_P T^*P & \longrightarrow & U \times_P U \\ \downarrow h_p J \times \text{id}_{T^*P} & & \downarrow h_p J \\ T^*P & \longrightarrow & P. \end{array}$$

By (4.8.7) and [Sai26, (1.39)], the support of the complex

$$(4.8.9) \quad \begin{aligned} & \mathcal{F}_{\psi, TP/P} (R(h_p \times \text{id}_{TP})_* J_{TP!} J_{TP}^* \nu_{i_p} \mathcal{H}_G) \\ &= \mathcal{F}_{\psi, TP/P} (R(h_p J \times \text{id}_{TP})_* J_{TP}^* \nu_{i_p} \mathcal{H}_G) \\ &= R(h_p J \times \text{id}_{T^*P})_* \mathcal{F}_{\psi, U \times_P U \times_P TP / U \times_P U} (J_{TP}^* \nu_{i_p} \mathcal{H}_G) \end{aligned}$$

is contained in the zero section $0_{T^*P} = T_P^*P \subset T^*P$.

We next consider the closed term. There are canonical isomorphisms

$$(4.8.10) \quad \begin{aligned} \mathbb{D}_Q G &= \mathbb{D}_Q((p^\vee)^* \text{Rad}(F)n-1) = (p^\vee)^! \mathbb{D}_{P^\vee} \text{Rad}(F)1-n \\ &= (p^\vee)^* \mathbb{D}_{P^\vee} \text{Rad}(F)[n-1], \end{aligned}$$

$$(4.8.11) \quad \mathcal{H}_G = \mathbb{D}_Q G \boxtimes G = (p^\vee \times p^\vee)^* (\mathbb{D}_{P^\vee} \text{Rad}(F) \boxtimes \text{Rad}(F))[2n-2](n-1).$$

By Lemma 3.19, the complex $\nu_{i_p} \mathcal{H}_G$ is W -invariant under the action of W described in Subsection 3.18. This W -invariant structure restricts to a $W|_Z$ -invariant structure on $I_{TP}^* \nu_{i_p} \mathcal{H}_G$. By Proposition 3.12 and Lemma 3.19,

$$(4.8.12) \quad \begin{aligned} & \text{supp } \mathcal{F}_{\psi, Z \times_P TP / Z} (I_{TP}^* \nu_{i_p} \mathcal{H}_G) \subset \ker(\beta^\vee|_Z) \\ &= \{(x, y_1, y_2, \xi) : \xi = 0 \text{ or } y_1 = y_2 = [H] \in Q_x, (y_1, y_2) \in Z, 0 \neq \xi \in T_x^*P, T_x H = \ker(\xi)\}. \end{aligned}$$

The following diagrams are cartesian:

$$(4.8.13) \quad \begin{array}{ccc} Z \times_P TP & \longrightarrow & Z \\ \downarrow h_p I \times \text{id}_{TP} & & \downarrow h_p I \\ TP & \longrightarrow & P, \end{array} \quad \begin{array}{ccc} Z \times_P T^*P & \longrightarrow & Z \\ \downarrow h_p I \times \text{id}_{T^*P} & & \downarrow h_p I \\ T^*P & \longrightarrow & P. \end{array}$$

By (4.8.12) and [Sai26, (1.39)], the support of the complex

$$(4.8.14) \quad \begin{aligned} & \mathcal{F}_{\psi, TP/P} (R(h_p \times \text{id}_{TP})_* I_{TP*} I_{TP}^* \nu_{i_p} \mathcal{H}_G) \\ &= \mathcal{F}_{\psi, TP/P} (R(h_p I \times \text{id}_{TP})_* I_{TP}^* \nu_{i_p} \mathcal{H}_G) \\ &= R(h_p I \times \text{id}_{T^*P})_* \mathcal{F}_{\psi, Z \times_P TP / Z} (I_{TP}^* \nu_{i_p} \mathcal{H}_G) \end{aligned}$$

is contained in

$$(4.8.15) \quad \begin{aligned} & (h_p I \times \text{id}_{T^*P})(\ker(\beta^\vee|_Z)) \\ &= \{(x, \xi) : \xi = 0 \text{ or } 0 \neq \xi \in T_x^*P, \exists (y_1, y_2) \in Z \text{ such that } y_1 = y_2 = [H], T_x H = \ker(\xi)\} \\ &= \{(x, \xi) : \xi \neq 0, (\rho(\xi), \rho(\xi)) \in Z\} \cup 0_{T^*P}(h_p(Z)), \end{aligned}$$

which is disjoint from $\rho^{-1}(U)$. $\square$

4.9. Proof of Theorem 4.2. By Lemma 4.5, it suffices to prove that, for every projective space P and every $F \in D_{\text{ctf}}^b(P, \Lambda)$, the complex F satisfies the projective singular support inclusion; that is,

$$(4.9.1) \quad \text{SS}_\mu(F)^\times = \text{SS}_\mu(F) \setminus T_P^*P \subset \rho^{-1}(E_p(p^{\vee*}\text{Rad}(F))).$$

Set $G = p^{\vee*}\text{Rad}(F)n-1$. Let $U \subseteq Q$ be an open subscheme disjoint from $E_p(p^{\vee*}\text{Rad}(F))$. Then $p|_U$ is universally locally acyclic relatively to $G|_U$. By Lemma 4.8, the complex

$$(4.9.2) \quad \mathcal{F}_{\psi, TP/P}(R(h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_G)$$

vanishes on $\rho^{-1}(U)$. By Lemma 3.5, the complex

$$(4.9.3) \quad \mu \mathcal{H}om_P(F, F) = \mathcal{F}_{\psi, TP/P} \nu_{\Delta_P} \mathcal{H}_F$$

is a retract of (4.9.2), and therefore also vanishes on $\rho^{-1}(U)$. Therefore $\text{SS}_\mu(F) = \text{supp} \mu \mathcal{H}om(F, F)$ is disjoint from $\rho^{-1}(U)$.

5. MICROLOCALIZATION ALONG CLOSED SUBSCHEMES

5.1. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes. Recall the Verdier specialization

$$(5.1.1) \quad \nu_{Z/X} : D_{\text{ctf}}^b(X, \Lambda) \rightarrow D_{\text{ctf}}^b(T_Z X, \Lambda).$$

As before, we suppress the η_0 factor. For $F \in D_{\text{ctf}}^b(X, \Lambda)$, the microlocalization of F along $Z \hookrightarrow X$ is the Fourier-Deligne transform

$$(5.1.2) \quad \mu_{Z/X}(F) = \mathcal{F}_{\psi, T_Z X/Z}(\nu_{Z/X}(F)) \in D_{\text{ctf}}^b(T_Z^* X, \Lambda).$$

Theorem 5.2. *Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes and let $F \in D_{\text{ctf}}^b(X, \Lambda)$. Then we have*

$$(5.2.1) \quad \text{supp } \mu_{Z/X}(F) \subseteq \text{SS}_\mu(F) \cap T_Z^* X \subseteq \text{SS}(F) \cap T_Z^* X.$$

The proof of Theorem 5.2 is given in Subsection 5.6.

5.3. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes. Let $T_Z X$ be the normal bundle of $i : Z \hookrightarrow X$. Then $T_Z^* X = (T_Z X)^\vee \subseteq T^* X|_Z$. Applying the construction of Subsection 3.16 to $i : Z \hookrightarrow X$ and $X \rightarrow \text{Spec}(k)$, we obtain

$$(5.3.1) \quad \tilde{\mu} : D_X(X \times X) \times_{X \times \mathbb{A}^1} D_Z(X) \rightarrow D_Z(X),$$

$$(5.3.2) \quad \mu : TX|_Z \times_Z T_Z X = TX \times_X T_Z X \rightarrow T_Z X.$$

We denote by $q : TX|_Z \rightarrow T_Z X$ the quotient bundle morphism. Unwinding the definitions, we see that for a geometric point $z \in Z$, the action

$$(5.3.3) \quad \mu_z : T_z X \times (T_Z X)_z \rightarrow (T_Z X)_z$$

is given as follows:

$$(5.3.4) \quad \mu_z = \text{add} \circ (q_z \times \text{id}) : T_z X \times T_z X / T_z Z \rightarrow T_z X / T_z Z.$$

We have a commutative diagram

$$(5.3.5) \quad \begin{array}{ccccc} D_Z(X) & \xrightarrow{D_{\text{id}}(\Delta)} & D_Z(X \times X) & \xrightarrow{D_{\text{id}}(\text{pr}_2)} & D_Z(X) \\ \downarrow D_i(\Delta) & \searrow \text{id} & \downarrow D_{\text{id}}(\text{pr}_1) & \nearrow D_i(\text{id}) & \downarrow \\ D_X(X \times X) & & D_Z(X) & & \end{array}$$

On the generic fiber, (5.3.5) is

$$(5.3.6) \quad \begin{array}{ccccc} X & \xrightarrow{\Delta} & X \times X & \xrightarrow{\text{pr}_2} & X \\ \downarrow \Delta & \searrow \text{id} & \downarrow \text{pr}_1 & \nearrow \text{id} & \downarrow \\ X \times X & & X & & \end{array}$$

and on the special fiber, (5.3.5) is

$$(5.3.7) \quad \begin{array}{ccccc} T_Z X & \xrightarrow{T_{\text{id}}(\Delta)} & T_Z(X \times X) & \xrightarrow{T_{\text{id}}(\text{pr}_2)} & T_Z X \\ \downarrow T_i(\Delta) & \searrow \text{id} & \downarrow T_{\text{id}}(\text{pr}_1) & \nearrow T_i(\text{id}) & \downarrow \\ T_X(X \times X) & & T_Z X & & \end{array}$$

By (3.16.2), the diagram (5.3.7) is isomorphic to the following diagram

$$(5.3.8) \quad \begin{array}{ccccc} T_Z X & \xrightarrow{(0_{TX}, \text{id})} & TX \times_X T_Z X & \xrightarrow{\mu} & T_Z X \\ \downarrow 0_{TX} & \searrow \text{id} & \downarrow \text{pr}_{T_Z X} & \nearrow \text{pr}_{TX} & \downarrow \\ TX & & TX & & \end{array}$$

Note that pr_{TX} factors as follows

$$(5.3.9) \quad \begin{array}{ccc} TX|_Z \times_Z T_Z X & \xlongequal{\quad} & TX \times_X T_Z X \\ \downarrow \text{pr}_{TX|_Z} & & \downarrow \text{pr}_{TX} \\ TX|_Z & \hookrightarrow & TX. \end{array}$$

Lemma 5.4. *Retain the notation of Subsection 5.3. For all $F \in D_{\text{ctf}}^b(X, \Lambda)$, the sheaf $\nu_{Z/X}(F) \in D_{\text{ctf}}^b(T_Z X, \Lambda)$ is a retract of*

$$(5.4.1) \quad \begin{aligned} & R\mu_!(\text{pr}_{TX}^* \nu \mathcal{H}om_X(F, F) \otimes \text{pr}_{T_Z X}^* \nu_{Z/X}(F)) \\ &= R\mu_!(\text{pr}_{TX|_Z}^*(\nu \mathcal{H}om_X(F, F)|_{TX|_Z}) \otimes \text{pr}_{T_Z X}^* \nu_{Z/X}(F)) \end{aligned}$$

Proof. We use the theory of cohomological correspondences over $S = (\mathbb{A}^1)_0^{\text{sh}}$ (c.f. Appendix A). We have the following commutative diagram in $\text{Corr}(\text{Sch}_{\mathbb{A}^1})^{\otimes}$

$$(5.4.2) \quad \begin{array}{ccc} (D_Z(X)) & \xrightarrow{\text{id}} & (D_Z(X)) \\ ([\Lambda], \text{id}) \downarrow & & \parallel \\ (D_X(X), D_Z(X)) & \xrightarrow{[\text{id}_!(D_i(\text{id}), \text{id})^*]} & (D_Z(X)) \\ ([D_{\text{id}}(\Delta)_!], \text{id}) \downarrow & & \parallel \\ (D_X(X \times X), D_Z(X)) & \xrightarrow{[D_{\text{id}}(\text{pr}_2)_!(D_i(\text{id}), D_{\text{id}}(\text{pr}_1))^*]} & (D_Z(X)), \end{array}$$

where the lower triangle is given by the following pullback diagram in $\text{Sch}_{\mathbb{A}^1}$ (c.f. Lemma 2.7)

$$(5.4.3) \quad \begin{array}{ccc} D_Z(X) & \xrightarrow{(D_i(\text{id}), \text{id})} & D_X(X) \times_{\mathbb{A}^1} D_Z(X) \\ \downarrow D_{\text{id}}(\Delta) & & \downarrow D_{\text{id}}(\Delta) \times \text{id} \\ D_Z(X \times X) & \xrightarrow{(D_i(\text{id}), D_{\text{id}}(\text{pr}_1))} & D_X(X \times X) \times_{\mathbb{A}^1} D_Z(X) \\ \downarrow D_{\text{id}}(\text{pr}_2) & & \\ D_Z(X) & & \end{array}$$

We then have a commutative diagram in $\text{CohCorr}_{S, \eta}^{\otimes}$

$$(5.4.4) \quad \begin{array}{ccc} (D_Z(X); F)_{\eta} & \xrightarrow{\text{id}} & (D_Z(X); F)_{\eta} \\ ([\Lambda], \text{id}; \text{id}, \text{id}) \downarrow & & \parallel \\ (D_X(X), D_Z(X); \Lambda, F)_{\eta} & & \\ ([D_{\text{id}}(\Delta)_!], \text{id}; \text{coev}, \text{id}) \downarrow & & \parallel \\ (D_X(X \times X), D_Z(X); \mathcal{H}_F, F)_{\eta} & \xrightarrow{([D_{\text{id}}(\text{pr}_2)_!(D_i(\text{id}), D_{\text{id}}(\text{pr}_1))^*]; \text{ev})} & (D_Z(X); F)_{\eta}, \end{array}$$

Applying the functor $\Psi^{\otimes} : \text{CohCorr}_{S, \eta}^{\otimes} \rightarrow \text{CohCorr}_{S, s}^{\otimes}$ to (5.4.4), we obtain a commutative diagram in $\text{CohCorr}_{S, s}^{\otimes}$:

$$(5.4.5) \quad \begin{array}{ccc} (D_Z(X); \nu_{Z/X}(F))_s & \xrightarrow{\text{id}} & (D_Z(X); \nu_{Z/X}(F))_s \\ \downarrow & & \parallel \\ (D_Z(X), D_Z(X); \Lambda, \nu_{Z/X}(F))_s & & \\ \downarrow & & \parallel \\ (D_X(X \times X), D_Z(X); \nu \mathcal{H}om_X(F, F), \nu_{Z/X}(F))_s & \longrightarrow & (D_Z(X); \nu_{Z/X}(F))_s, \end{array}$$

whose bottom morphism factors through the cocartesian edge

$$(5.4.6) \quad \begin{array}{c} (D_X(X \times X), D_Z(X); \nu_{X/X \times X} \mathcal{H}_F, \nu_{Z/X}(F))_s \\ \rightarrow (D_Z(X); R\mu_!(\text{pr}_{TX}^* \nu \mathcal{H}om_X(F, F) \otimes \text{pr}_{TZ}^* \nu_{Z/X}(F)))_s. \end{array}$$

□

Corollary 5.5. *Consider the notation in 5.3. For all $F \in D_{\text{ctf}}^b(X, \Lambda)$ the sheaf*

$$(5.5.1) \quad \mu_{Z/X}(F) \in D_{\text{ctf}}^b(T_Z^* X, \Lambda)$$

is a retract of

$$(5.5.2) \quad (q^{\vee})^*(\mu \mathcal{H}om_X(F, F)|_{T^* X|_Z}) \otimes \mu_{Z/X}(F).$$

Note that this defines a canonical morphism

$$(5.5.3) \quad (q^\vee)^*(\mu\mathcal{H}om_X(F, F)|_{T^*X|_Z}) \otimes \mu_{Z/X}(F) \rightarrow \mu_{Z/X}(F).$$

Proof. Applying Lemma 3.15 to $q : TX|_Z \rightarrow T_ZX$ and the following diagram

$$(5.5.4) \quad \begin{array}{ccc} & TX|_Z \times_Z T_ZX & \xrightarrow{\mu} T_ZX \\ & \swarrow \text{pr}_{TX|_Z} & \downarrow \text{pr}_{T_ZX} \\ TX|_Z & & T_ZX, \end{array}$$

we see that (5.5.2) is the Fourier transform of (5.4.1). $\square$

5.6. Proof of 5.2. Let $U \subset T^*X$ be an open subset where $\mu\mathcal{H}om_X(F, F)$ vanishes. By 5.5, $\mu_{Z/X}(F)$ is a retract of

$$(5.6.1) \quad (q^\vee)^*(\mu\mathcal{H}om_X(F, F)|_{T^*X|_Z}) \otimes \mu_{Z/X}(F)$$

which vanishes on $q^{\vee-1}(U) = U \cap T_Z^*X$ (note that $q^\vee : T_Z^*X \hookrightarrow T^*X|_Z$). It follows that

$$(5.6.2) \quad \text{supp} \mu_{Z/X}(F) \subset \text{supp} \mu\mathcal{H}om_X(F, F) \cap T_Z^*X = \text{SS}_\mu(F) \cap T_Z^*X \subseteq \text{SS}(F) \cap T_Z^*X,$$

where the last inclusion follows from (4.2.1).

6. SAITO'S CONJECTURE ON CHARACTERISTIC CLASSES

In this section, we apply Theorem 1.3 to characteristic classes. We first prove the following comparison.

Corollary 6.1. *Let Y be a smooth connected scheme and $K \in D_{\text{ctf}}^b(Y, \Lambda)$. Consider Saito's microlocal characteristic cycle [Sai26, Definition 2.1.4]*

$$(6.1.1) \quad \text{CC}_\mu(K) \in H_{\text{SS}_\mu(K)}^0(T^*Y, e_Y^\vee \mathcal{K}_{Y/k}),$$

where $e_Y^\vee : T^*Y \rightarrow Y$ is the projection. We have

$$(6.1.2) \quad \text{CC}_\mu(K) = \text{cl}(\text{CC}(K)).$$

Proof. By Theorem 4.2, we have $\dim \text{SS}_\mu(K) \leq \dim Y$. Now (6.1.2) follows from [Sai26, Proposition 2.5.2]. $\square$

6.2. Let Y be a smooth scheme and $f : Y \rightarrow S$ a morphism to a smooth curve. Let $s \in S$, and let t be a local parameter at s . After shrinking S around s , we may assume that dt is nowhere zero and defines a section $dt : S \rightarrow T^*S$. Let $\theta_{f,t} : Y \rightarrow T^*Y$ be the composition

$$(6.2.1) \quad Y \xrightarrow{dt} Y \times_S T^*S \xrightarrow{df} T^*Y.$$

We impose the following condition:

$$(6.2.2) \quad \theta_{f,t}^{-1} \text{SS}_\mu(K) \subset Y_s.$$

Under the condition (6.2.2), we have

$$(6.2.3) \quad \theta_{f,t}^* \text{CC}_\mu(K) \in H_{Y_s}^0(Y, \mathcal{K}_{Y/k}).$$

Proposition 6.3. *Let Y be a smooth scheme, let S be a smooth connected curve, and let $f : Y \rightarrow S$ be a proper morphism. Suppose that f is universally locally acyclic relatively to $K \in D_{\text{ctf}}^b(Y, \Lambda)$ over $S \setminus \{s\}$ and that (6.2.2) holds. Then we have*

$$(6.3.1) \quad f_*(\theta_{f,t}^* \text{CC}_\mu(K)) = -a_s(Rf_*K)[s] \quad \text{in} \quad H_s^0(S, \mathcal{K}_{S/k}),$$

where a_s is the Artin conductor.

Proof. We work over the henselization $S_{(s)}$ and take cohomology with support on the closed fiber. Consider the correspondence

$$(6.3.2) \quad T^*Y \xleftarrow{df} Y \times_S T^*S \xrightarrow{\text{pr}} T^*S.$$

Put $C_S = f_* \text{SS}_\mu(K) \cup \text{SS}(Rf_*K)$. By the assumption, we have $dt^{-1}(C_S) \subseteq \{s\}$. By condition (6.2.2), We have a commutative diagram

$$(6.3.3) \quad \begin{array}{ccc} H_{\text{SS}_\mu(K)}^0(T^*Y, e_Y^{\vee*} \mathcal{K}_{Y/k}) & \xrightarrow{f_!} & H_{C_S}^0(T^*S, e_S^{\vee*} \mathcal{K}_{S/k}) \\ \theta_{f,t}^* \downarrow & & \downarrow \theta_{\text{id},t}^* \\ H_{Y_s}^0(Y, \mathcal{K}_{Y/k}) & \xrightarrow{f_*} & H_s^0(S, \mathcal{K}_{S/k}). \end{array}$$

By [Sai26, Proposition 2.3.3], we have

$$(6.3.4) \quad f_! \text{CC}_\mu(K) = \text{CC}_\mu(Rf_*K).$$

On the curve S , this class is the cycle class of $\text{CC}(Rf_*K)$ by [Sai26, Proposition 2.5.1]. The coefficient of T_s^*S in $\text{CC}(Rf_*K)$ is $-a_s(Rf_*K)$. Since dt is disjoint from the zero section and meets T_s^*S transversally with multiplicity one, it follows that

$$(6.3.5) \quad f_* \theta_{f,t}^*(\text{CC}_\mu(K)) = -a_s(Rf_*K)[s].$$

This finishes the proof. $\square$

6.4. Let $i : Z \hookrightarrow M$ be a closed immersion with M smooth and connected. By Grothendieck duality, we have $\mathcal{K}_{Z/k} = i^! \mathcal{K}_{M/k}$. Excision gives an isomorphism

$$(6.4.1) \quad \iota_Z : H^0(Z, \mathcal{K}_{Z/k}) \xrightarrow{\sim} H_Z^0(M, \mathcal{K}_{M/k}).$$

Let $0_M : M \hookrightarrow T^*M$ be the zero section. Let $F \in D_{\text{ctf}}^b(Z, \Lambda)$ and $G = i_* F$. Since both $\text{SS}_\mu(G)$ and $\text{SS}(G)$ lie over Z , pullback by the zero-section defines a morphism

$$(6.4.2) \quad 0_{M,Z}^* : H_{\text{SS}(G)}^0(T^*M, e_M^{\vee*} \mathcal{K}_{M/k}) \longrightarrow H_Z^0(M, \mathcal{K}_{M/k}).$$

Lemma 6.5. *With the notation of Subsection 6.4, we have*

$$(6.5.1) \quad 0_{M,Z}^* \text{CC}_\mu(i_* F) = \iota_Z(C_{Z/k}(F)).$$

Proof. Put $j : M \setminus Z \hookrightarrow M$. By the localization triangle and duality, we have canonical isomorphisms

$$(6.5.2) \quad R\Gamma_Z \mathcal{K}_{M/k} := \text{fib}(\mathcal{K}_{M/k} \rightarrow Rj_* j^* \mathcal{K}_{M/k}) \xrightarrow{\sim} i_* i^! \mathcal{K}_{M/k} = i_* \mathcal{K}_{Z/k}, \quad \mathbb{D}_M(i_* F) \xrightarrow{\sim} i_* \mathbb{D}_Z F.$$

Therefore the evaluation morphism for $G = i_* F$ has a canonical factorization with support:

$$(6.5.3) \quad \mathbb{D}_M G \otimes^L G \longrightarrow i_*(\mathbb{D}_Z F \otimes^L F) \xrightarrow{i_* \text{ev}_F} i_* \mathcal{K}_{Z/k} \xrightarrow{\sim} R\Gamma_Z \mathcal{K}_{M/k} \longrightarrow \mathcal{K}_{M/k}.$$

The first morphism is induced by the projection formula. After applying i^* , it becomes the canonical tensor-product morphism on Z and hence is an isomorphism.

Consider $\mathcal{H}_G = R\mathcal{H}om(\text{pr}_1^* G, \text{pr}_2^* G)$ and $\mathcal{H}_F = R\mathcal{H}om(\text{pr}_1^* F, \text{pr}_2^* F)$. By proper base change, we have canonical isomorphisms

$$(6.5.4) \quad \mathcal{H}_G = R(i \times i)_* \mathcal{H}_F, \quad \Delta_M^! \mathcal{H}_G = i_* \Delta_Z^! \mathcal{H}_F, \quad \Delta_M^* \mathcal{H}_G = i_* \Delta_Z^* \mathcal{H}_F.$$

Under these isomorphisms, the full faithfulness of i_* identifies id_G with $i_*(\text{id}_F)$, and the projection formula identifies the evaluation morphisms. We therefore obtain a commutative diagram in

$D_{\text{ctf}}^b(M, \Lambda)$:

$$(6.5.5) \quad \begin{array}{ccccccc} \Lambda_M & \xrightarrow{1_G} & \Delta_M^! \mathcal{H}_G & \longrightarrow & \Delta_M^* \mathcal{H}_G & \xrightarrow{\text{ev}_{G,Z}} & R\Gamma_Z \mathcal{K}_{M/k} \\ \downarrow & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ i_* \Lambda_Z & \xrightarrow{i_* 1_F} & i_* \Delta_Z^! \mathcal{H}_F & \longrightarrow & i_* \Delta_Z^* \mathcal{H}_F & \xrightarrow{i_* \text{ev}_F} & i_* \mathcal{K}_{Z/k}, \end{array}$$

where $\text{ev}_{G,Z}$ is the supported evaluation morphism

$$(6.5.6) \quad \text{ev}_{G,Z} : \Delta_M^* \mathcal{H}_G \simeq \mathbb{D}_M G \otimes^L G \xrightarrow{(6.5.3)} R\Gamma_Z \mathcal{K}_{M/k}.$$

We compare (6.5.5) with microlocalization. Note that $\nu \mathcal{H}_G$ is monodromic by [Ver83, Section 8 (SP1)]. We have isomorphisms

$$(6.5.7) \quad 0_M^* \mu \mathcal{H}om(G, G) \simeq e_! \nu \mathcal{H}_G \simeq 0_{TM}^! \nu \mathcal{H}_G \simeq \Delta_M^! \mathcal{H}_G,$$

where the first isomorphism follows from [Sai26, Lemma 1.1.2.1] and the second follows from [Ver83, Lemma 6.1] for monodromic sheaves, and the third isomorphism follows from [Sai26, Proposition 1.2.8]. From (6.5.5), we obtain a commutative diagram with support:

$$(6.5.8) \quad \begin{array}{ccccc} \Lambda_M & \longrightarrow & 0_M^* \mu \mathcal{H}om(G, G) & \xrightarrow{\text{ev}'_{\mu,Z}} & R\Gamma_Z \mathcal{K}_{M/k} \\ \parallel & & \downarrow \sigma_G & & \downarrow \\ \Lambda_M & \xrightarrow{1_G} & \Delta_M^! \mathcal{H}_G & \longrightarrow & \mathcal{K}_{M/k}, \end{array}$$

where $\text{ev}'_{\mu,Z}$ is the composition

$$(6.5.9) \quad 0_M^* \mu \mathcal{H}om(G, G) \simeq \Delta_M^! \mathcal{H}_G \rightarrow \Delta_M^* \mathcal{H}_G \xrightarrow{\text{ev}_{G,Z}} R\Gamma_Z \mathcal{K}_{M/k}.$$

The lower composite is the upper row of (6.5.5) followed by the forget-support morphism. The morphism σ_G is the composite of Fourier base change and the zero-section unit for specialization. The left square is the Fourier–specialization unit square, and the right square follows from the naturality of evaluation.

Combining (6.5.5) and (6.5.8), we obtain the factorization

$$(6.5.10) \quad \Lambda_M \longrightarrow i_* \Lambda_Z \xrightarrow{i_* C_{Z/k}(F)} i_* \mathcal{K}_{Z/k} = R\Gamma_Z \mathcal{K}_{M/k} \longrightarrow \mathcal{K}_{M/k}.$$

By excision, the part of (6.5.10) ending in $R\Gamma_Z \mathcal{K}_{M/k}$ represents $\iota_Z(C_{Z/k}(F))$. On the other hand, (6.5.8) identifies it with $0_{M,Z}^* \text{CC}_\mu(G)$. This proves (6.5.1). $\square$

Lemma 6.6. *Consider the notation in 6.4. Let $\text{cl}_M : CH_{\dim M}(\text{SS}(i_* F)) \rightarrow H_{\text{SS}(i_* F)}^0(T^* M, e_M^{\vee*} \mathcal{K}_{M/k})$ be the cycle-class map with support. Then we have*

$$(6.6.1) \quad 0_{M,Z}^* \text{cl}_M(\text{CC}(i_* F)) = \iota_Z(\text{cl}_Z(cc_{Z,0}(F))).$$

Proof. The cycle $\text{CC}(i_* F)$ is a cycle on $T^* M$ whose base is contained in Z . Since the zero section is a regular immersion, the compatibility of the cycle-class map with refined Gysin pullback gives a commutative diagram

$$(6.6.2) \quad \begin{array}{ccc} CH_{\dim M}(\text{SS}(i_* F)) & \xrightarrow{\text{cl}_M} & H_{\text{SS}(i_* F)}^0(T^* M, e_M^{\vee*} \mathcal{K}_{M/k}) \\ 0_M^! \downarrow & & \downarrow 0_{M,Z}^* \\ CH_0(Z) & \xrightarrow{\iota_Z \circ \text{cl}_Z} & H_Z^0(M, \mathcal{K}_{M/k}). \end{array}$$

The lower horizontal morphism is the cycle-class map followed by the excision isomorphism, since the intersection is supported on Z . Applying the diagram to $\text{CC}(i_*F)$, we obtain

$$(6.6.3) \quad 0_{M,Z}^* \text{cl}_M(\text{CC}(i_*F)) = \iota_Z \text{cl}_Z(0_M^! \text{CC}(i_*F)) = \iota_Z \text{cl}_Z(cc_{Z,0}(F)),$$

where the second equality follows from [Sai17, Definition 6.7.2 and Lemma 6.9.1]. $\square$

Theorem 6.7 ([Sai17, Conjecture 6.8]). *Let Z be a k -scheme admitting a closed immersion into a smooth k -scheme. For every $F \in D_{\text{ctf}}^b(Z, \Lambda)$, we have*

$$(6.7.1) \quad C_{Z/k}(F) = \text{cl}_Z(cc_{Z,0}(F)) \quad \text{in } H^0(Z, \mathcal{K}_{Z/k}).$$

Proof. Take a closed immersion $i : Z \hookrightarrow M$ with M smooth, and put $G = i_*F$. By Theorem 1.3, we have $\text{SS}_\mu(G) \subset \text{SS}(G)$. By [Bei16, Theorem 1.2], $\dim \text{SS}(G) \leq \dim M$. Hence $\dim \text{SS}_\mu(G) \leq \dim M$. Applying [Sai26, Proposition 2.5.2], we obtain

$$(6.7.2) \quad \text{CC}_\mu(G) = \text{cl}_M(\text{CC}(G)) \quad \text{in } H_{\text{SS}(G)}^0(T^*M, e_M^{\vee*} \mathcal{K}_{M/k}),$$

where, by abuse of notation, $\text{CC}_\mu(G)$ also denotes its image under the map $H_{\text{SS}_\mu(G)}^0(T^*M, e_M^{\vee*} \mathcal{K}_{M/k}) \rightarrow H_{\text{SS}(G)}^0(T^*M, e_M^{\vee*} \mathcal{K}_{M/k})$. Apply $0_{M,Z}^*$ to (6.7.2). By Lemma 6.5 and Lemma 6.6, the two sides are, respectively,

$$(6.7.3) \quad \iota_Z(C_{Z/k}(F)) \quad \text{and} \quad \iota_Z(\text{cl}_Z(cc_{Z,0}(F))).$$

Since ι_Z is an isomorphism, we obtain (6.7.1). $\square$

7. EXTERNAL PRODUCTS

7.1. Let X and Y be smooth k -schemes. Let $F \in D_{\text{ctf}}^b(X, \Lambda)$ and $G \in D_{\text{ctf}}^b(Y, \Lambda)$. We identify

$$(7.1.1) \quad T^*(X \times Y) = T^*X \times T^*Y$$

by the canonical decomposition $\Omega_{X \times Y/k}^1 = \text{pr}_X^* \Omega_{X/k}^1 \oplus \text{pr}_Y^* \Omega_{Y/k}^1$.

For cohomology classes with support, $\alpha \boxtimes \beta$ denotes the cohomological external product followed by the canonical isomorphism (cf. [YZ25, (3.2)])

$$(7.1.2) \quad \mathcal{K}_{X/k} \boxtimes \mathcal{K}_{Y/k} \xrightarrow{\sim} \mathcal{K}_{X \times Y/k}.$$

Lemma 7.2. *Let $\mathcal{H}_F = R\mathcal{H}om(p_{1,X}^* F, p_{2,X}^! F)$ and $\mathcal{H}_G = R\mathcal{H}om(p_{1,Y}^* G, p_{2,Y}^! G)$. There are canonical isomorphisms*

$$(7.2.1) \quad \mathcal{H}_{F \boxtimes G} \simeq \mathcal{H}_F \boxtimes \mathcal{H}_G,$$

$$(7.2.2) \quad \nu_{\Delta_{X \times Y}}(\mathcal{H}_{F \boxtimes G}) \simeq \nu_{\Delta_X}(\mathcal{H}_F) \boxtimes_{\eta_0} \nu_{\Delta_Y}(\mathcal{H}_G).$$

Proof. By [LZ22, Lemma 2.11(a)], we have natural isomorphisms

$$(7.2.3) \quad \begin{aligned} \mathcal{H}_{F \boxtimes G} &\simeq \mathbb{D}_{X \times Y}(F \boxtimes G) \boxtimes (F \boxtimes G) \\ &\simeq (\mathbb{D}_X(F) \boxtimes \mathbb{D}_Y(G)) \boxtimes (F \boxtimes G) \simeq \mathcal{H}_F \boxtimes \mathcal{H}_G. \end{aligned}$$

Applying Lemma 2.8 to Δ_X and Δ_Y , we obtain a natural isomorphism

$$(7.2.4) \quad \text{D}_{X \times Y}(X \times X \times Y \times Y) \simeq \text{D}_X(X \times X) \times_{\mathbb{A}^1} \text{D}_Y(Y \times Y).$$

Applying nearby cycles to (7.2.1) and [LZ22, Proposition 3.1], we obtain (7.2.2). $\square$

Theorem 7.3 (Microlocal Künneth formula). *After the identification (7.1.1), there is a canonical isomorphism*

$$(7.3.1) \quad \mu\mathcal{H}om(F \boxtimes G, F \boxtimes G) \xrightarrow{\sim} \mu\mathcal{H}om(F, F) \boxtimes \mu\mathcal{H}om(G, G).$$

Consequently,

$$(7.3.2) \quad \text{SS}_\mu(F \boxtimes G) = \text{SS}_\mu(F) \times \text{SS}_\mu(G),$$

$$(7.3.3) \quad \mathrm{CC}_\mu(F \boxtimes G) = \mathrm{CC}_\mu(F) \boxtimes \mathrm{CC}_\mu(G).$$

Proof. By Lemma 7.2 and Lemma 3.14, we have

$$(7.3.4) \quad \begin{aligned} \mu\mathrm{Hom}(F \boxtimes G, F \boxtimes G) &= \mathfrak{F}_{\psi, T(X \times Y)/(X \times Y)} \nu_{\Delta_{X \times Y}}(\mathcal{H}_{F \boxtimes G}) \\ &\simeq \mathfrak{F}_{\psi, TX \times TY/(X \times Y)} (\nu_{\Delta_X}(\mathcal{H}_F) \boxtimes_{\eta_0} \nu_{\Delta_Y}(\mathcal{H}_G)) \\ &\simeq \mathfrak{F}_{\psi, TX/X} \nu_{\Delta_X}(\mathcal{H}_F) \boxtimes_{\eta_0} \mathfrak{F}_{\psi, TY/Y} \nu_{\Delta_Y}(\mathcal{H}_G) \\ &= \mu\mathrm{Hom}(F, F) \boxtimes_{\eta_0} \mu\mathrm{Hom}(G, G). \end{aligned}$$

On supports we have

$$(7.3.5) \quad \begin{aligned} \mathrm{SS}_\mu(F \boxtimes G) &= \mathrm{pr}_{T^*(X \times Y)} \mathrm{supp}(\mu\mathrm{Hom}(F, F) \boxtimes_{\eta_0} \mu\mathrm{Hom}(G, G)) \\ &= \mathrm{pr}_{T^*X} \mathrm{supp} \mu\mathrm{Hom}(F, F) \times \mathrm{pr}_{T^*Y} \mathrm{supp} \mu\mathrm{Hom}(G, G). \end{aligned}$$

This is (7.3.2).

It remains to prove (7.3.3). Consider the kernel $\mathcal{H}_F$, the cohomological characteristic class of F is represented by the composition

$$(7.3.6) \quad \Lambda_X \xrightarrow{1_F} \Delta_X^! \mathcal{H}_F \longrightarrow \Delta_X^* \mathcal{H}_F \xrightarrow{\mathrm{ev}_F} \mathcal{K}_{X/k}.$$

We have the analogous composites for G and $F \boxtimes G$.

The compatibility of (7.2.1) with the identity morphisms gives a commutative diagram

$$(7.3.7) \quad \begin{array}{ccc} \Lambda_X \boxtimes \Lambda_Y & \xrightarrow{1_F \boxtimes 1_G} & \Delta_X^! \mathcal{H}_F \boxtimes \Delta_Y^! \mathcal{H}_G \\ \sim \downarrow & & \downarrow \sim \\ \Lambda_{X \times Y} & \xrightarrow{1_{F \boxtimes G}} & \Delta_{X \times Y}^! \mathcal{H}_{F \boxtimes G} \end{array}$$

The canonical morphisms $\Delta^! \rightarrow \Delta^*$ give a commutative diagram

$$(7.3.8) \quad \begin{array}{ccc} \Delta_X^! \mathcal{H}_F \boxtimes \Delta_Y^! \mathcal{H}_G & \longrightarrow & \Delta_X^* \mathcal{H}_F \boxtimes \Delta_Y^* \mathcal{H}_G \\ \sim \downarrow & & \downarrow \sim \\ \Delta_{X \times Y}^! \mathcal{H}_{F \boxtimes G} & \longrightarrow & \Delta_{X \times Y}^* \mathcal{H}_{F \boxtimes G}. \end{array}$$

Its commutativity follows from the naturality of $\Delta^! \rightarrow \Delta^*$ and its compatibility with external products. The evaluation morphisms give a commutative diagram

$$(7.3.9) \quad \begin{array}{ccc} \Delta_X^* \mathcal{H}_F \boxtimes \Delta_Y^* \mathcal{H}_G & \xrightarrow{\mathrm{ev}_F \boxtimes \mathrm{ev}_G} & \mathcal{K}_{X/k} \boxtimes \mathcal{K}_{Y/k} \\ \sim \downarrow & & \downarrow \sim \\ \Delta_{X \times Y}^* \mathcal{H}_{F \boxtimes G} & \xrightarrow{\mathrm{ev}_{F \boxtimes G}} & \mathcal{K}_{X \times Y/k}. \end{array}$$

The vertical morphisms are the external-product isomorphisms and (7.1.2). Diagram (7.3.7) follows from $\mathrm{id}_{F \boxtimes G} = \mathrm{id}_F \boxtimes \mathrm{id}_G$. To verify (7.3.9), use $\mathcal{H}_F \simeq \mathbb{D}_X F \boxtimes F$ and $\mathcal{H}_G \simeq \mathbb{D}_Y G \boxtimes G$. Under $\mathbb{D}_{X \times Y}(F \boxtimes G) = \mathbb{D}_X F \boxtimes \mathbb{D}_Y G$, the evaluation $\mathrm{ev}_{F \boxtimes G}$ is the composite of the symmetry isomorphism

$$(7.3.10) \quad (\mathbb{D}_X F \boxtimes \mathbb{D}_Y G) \otimes (F \boxtimes G) \xrightarrow{\sim} (\mathbb{D}_X F \otimes F) \boxtimes (\mathbb{D}_Y G \otimes G)$$

and $\mathrm{ev}_F \boxtimes \mathrm{ev}_G$, followed by (7.1.2). Hence (7.3.9) is commutative.

By adjunction, (7.3.6) defines a correspondence

$$(7.3.11) \quad \Delta_X^! \Lambda_X \longrightarrow \mathcal{H}_F \longrightarrow \Delta_{X*} \mathcal{K}_{X/k}.$$

Let $0_X : X \hookrightarrow TX$ be the zero section. Applying Verdier specialization to (7.3.11), we obtain

$$(7.3.12) \quad 0_{X!} \Lambda_X \longrightarrow \nu_{\Delta_X}(\mathcal{H}_F) \longrightarrow 0_{X*} \mathcal{K}_{X/k}.$$

Here we use the canonical isomorphisms $\nu_{\Delta_X}(\Delta_X! \Lambda_X) = 0_{X!} \Lambda_X$ and $\nu_{\Delta_X}(\Delta_X^* \mathcal{K}_{X/k}) = 0_{X^*} \mathcal{K}_{X/k}$. Applying the Fourier-Deligne transformation and the canonical isomorphisms

$$(7.3.13) \quad \mathfrak{F}_{\psi, TX/X}(0_{X!} \Lambda_X) = \Lambda_{T^*X}, \quad \mathfrak{F}_{\psi, TX/X}(0_{X^*} \mathcal{K}_{X/k}) = e_X^{\vee*} \mathcal{K}_{X/k}$$

we obtain morphisms

$$(7.3.14) \quad \Lambda_{T^*X} \xrightarrow{u_F} \mu\mathcal{H}om(F, F) \xrightarrow{v_F} e_X^{\vee*} \mathcal{K}_{X/k}.$$

These are the two structural morphisms defining $\mathrm{CC}_\mu(F)$.

The three commutative diagrams (7.3.7), (7.3.8), and (7.3.9) can be specialized simultaneously. In fact, the Künneth morphisms are the structural isomorphisms of the symmetric monoidal nearby-cycle functor [LZ22, Construction 3.3]. The naturality of the isomorphism (3.14.1) in Lemma 3.14 preserves the three resulting morphisms. We obtain a commutative diagram

$$(7.3.15) \quad \begin{array}{ccccc} \Lambda_{T^*X} \boxtimes \Lambda_{T^*Y} & \xrightarrow{u_F \boxtimes u_G} & \mu\mathcal{H}om(F, F) \boxtimes \mu\mathcal{H}om(G, G) & \xrightarrow{v_F \boxtimes v_G} & e_X^{\vee*} \mathcal{K}_{X/k} \boxtimes e_Y^{\vee*} \mathcal{K}_{Y/k} \\ \sim \downarrow & & \sim \downarrow & & \downarrow \sim \\ \Lambda_{T^*(X \times Y)} & \xrightarrow{u_F \boxtimes u_G} & \mu\mathcal{H}om(F \boxtimes G, F \boxtimes G) & \xrightarrow{v_F \boxtimes v_G} & e_{X \times Y}^{\vee*} \mathcal{K}_{X \times Y/k}. \end{array}$$

The left vertical morphism is induced by the external product of constant sheaves, and the right one is induced by (7.1.2). Thus (7.3.15) is compatible with both structural morphisms defining the microlocal characteristic cycle.

We now pass to cohomology with support. Let $i_F : \mathrm{SS}_\mu(F) \hookrightarrow T^*X$ and $i_G : \mathrm{SS}_\mu(G) \hookrightarrow T^*Y$ be the closed immersions. By (7.3.2), the microlocal singular support of $F \boxtimes G$ is $\mathrm{SS}_\mu(F) \times \mathrm{SS}_\mu(G)$. By definition, $\mathrm{CC}_\mu(F)$ is represented on $\mathrm{SS}_\mu(F)$ by the composite

$$(7.3.16) \quad \Lambda_{C_F} \longrightarrow i_F^* \mu\mathcal{H}om(F, F) \longrightarrow i_F^! e_X^{\vee*} \mathcal{K}_{X/k},$$

obtained from (7.3.14) by adjunction. By (7.3.15), the external product of (7.3.16) and its analogue for G is the corresponding composite on $\mathrm{SS}_\mu(F) \times \mathrm{SS}_\mu(G)$. Here we use the canonical isomorphism

$$(7.3.17) \quad (i_F^! e_X^{\vee*} \mathcal{K}_{X/k}) \boxtimes (i_G^! e_Y^{\vee*} \mathcal{K}_{Y/k}) \xrightarrow{\sim} (i_F \times i_G)^! e_{X \times Y}^{\vee*} \mathcal{K}_{X \times Y/k}.$$

Taking H^0 , the resulting class is $\mathrm{CC}_\mu(F) \boxtimes \mathrm{CC}_\mu(G)$. By the lower row of (7.3.15), it is also $\mathrm{CC}_\mu(F \boxtimes G)$. This proves (7.3.3). $\square$

8. EQUALITY FOR PERVERSE SHEAVES

8.1. Let $E/\mathbf{Q}_\ell$ be a finite extension, let $\mathcal{O}_E$ be the ring of integers in E , and let π be a uniformizer. Let Y be a scheme of finite type over k . Let $D_{\mathrm{cons}}(Y_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ be the stable ∞ -category of constructible pro-étale sheaves on Y with $\mathcal{O}_E$ -coefficients, which is derived π -complete. By [HRS23, Theorem 1.6 (1) and (2)], there is an equivalence of stable ∞ -categories

$$(8.1.1) \quad D_{\mathrm{cons}}(Y_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E) = \varprojlim_m D_{\mathrm{ctf}}^b(Y, \mathcal{O}_E/\pi^m).$$

By [HS23, Corollary 2.4], every constructible E -sheaf has an integral model. By [Bar24, Lemma 2.1], reduction modulo π is conservative on derived complete constructible objects.

The operations used to form the diagonal endomorphism kernel, Verdier specialization, and Fourier-Deligne transform preserve constructibility and are compatible with reduction. Hence, for an integral model F_0 of $F \in D_{\mathrm{cons}}(Y_{\mathrm{pro\acute{e}t}}, E)$ and $F_m = F_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m$, there are canonical isomorphisms

$$(8.1.2) \quad \mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0) \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m \xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}_E/\pi^m}(F_m, F_m),$$

$$(8.1.3) \quad \mu\mathcal{H}om_E(F, F) \xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0) \otimes_{\mathcal{O}_E} E.$$

The right side of (8.1.3) is independent of F_0 because it realizes the intrinsically defined object on $Y_{\text{proét}}$.

By [Bar24, Proposition 3.6], universal local acyclicity on the pro-étale site is equivalent to dualizability in the relative correspondence category. It is preserved by smooth pullback and proper pushforward. Moreover, Beilinson's singular support and its projective Radon description hold for constructible E -sheaves (cf. [Bar24]). Following [UYZ20], the characteristic cycle of an ℓ -adic sheaf F is defined by

$$(8.1.4) \quad CC(F) := CC(F_1),$$

where $F_1 = F_0 \otimes^L \mathcal{O}_E/\pi$ for any integral model F_0 of F .

Theorem 8.2. *Let X be a smooth k -scheme, and let $F \in D_{\text{cons}}(X_{\text{proét}}, E)$. Then*

$$(8.2.1) \quad \text{SS}_\mu(F) \subseteq \text{SS}(F).$$

Proof. The assertion is clear if $F = 0$. By [Bar24, Theorem 1.5(v)], there exists an integral model $F_0 \in D_{\text{cons}}(X_{\text{proét}}, \mathcal{O}_E)$ of F such that

$$(8.2.2) \quad F_0 \otimes_{\mathcal{O}_E} E \simeq F, \quad \text{SS}(F_0) = \text{SS}(F).$$

Set $F_1 = F_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi$. Under the equivalence between constructible sheaves on the pro-étale and the étale sites for finite coefficients [HRS23, Theorem 1.6.(1)], we have an equality by [Sai17, Corollary 4.5.1]

$$(8.2.3) \quad \text{SS}(F_1) = \text{SS}(F_0) \stackrel{(8.2.2)}{=} \text{SS}(F).$$

We next compare the microlocal Hom complexes. As in the preceding sections, we suppress the harmless η_0 -factor. The isomorphisms (8.1.2) and (8.1.3) give

$$(8.2.4) \quad \begin{aligned} \mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0) \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi &\xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}_E/\pi}(F_1, F_1), \\ \mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0) \otimes_{\mathcal{O}_E} E &\xrightarrow{\sim} \mu\mathcal{H}om_E(F, F). \end{aligned}$$

Let us recall why these compatibilities hold. The diagonal endomorphism kernel can be written as

$$(8.2.5) \quad \mathcal{H}_{\mathcal{O}_E}(F_0, F_0) \simeq \mathbb{D}_{\mathcal{O}_E}(F_0) \boxtimes F_0.$$

For constructible complexes, Verdier duality and external products commute with reduction modulo π and with rationalization. Verdier specialization along Δ_X has the same coefficient compatibility. Finally, the Fourier–Deligne transform is constructed from pullback, tensor product with the Artin–Schreier sheaf, and compactly supported pushforward, all of which commute with these changes of coefficients. This proves (8.2.4).

We claim that

$$(8.2.6) \quad \text{SS}_\mu(F) \subseteq \text{SS}_\mu(F_1).$$

Let $U = T^*X \setminus \text{supp}(\mu\mathcal{H}om_{\mathcal{O}_E/\pi}(F_1, F_1))$. By definition, $\mu\mathcal{H}om_{\mathcal{O}_E/\pi}(F_1, F_1)|_U = 0$. Using the first isomorphism in (8.2.4), we obtain $\mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0)|_U \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi = 0$. The complex

$$(8.2.7) \quad N = \mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0)|_U$$

is constructible and derived π -complete. Derived Nakayama therefore gives $N = 0$. More explicitly, the distinguished triangle

$$(8.2.8) \quad N \xrightarrow{\pi} N \longrightarrow N \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi \longrightarrow$$

shows that multiplication by π on N is an isomorphism. Hence $N \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m = 0$ for every $m \geq 1$. Since N is derived π -complete, we have $N \simeq R\varprojlim_m (N \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m) = 0$. This is precisely the uniformizer version of the derived Nakayama argument in [Bar24, Lemma 2.1].

Rationalizing and using the second isomorphism in (8.2.4), we obtain

$$(8.2.9) \quad \mu\mathcal{H}om_E(F, F)|_U \simeq \mu\mathcal{H}om_{\mathcal{O}_E}(F_0, F_0)|_U \otimes_{\mathcal{O}_E} E = 0.$$

It follows that

$$(8.2.10) \quad \text{supp}(\mu\mathcal{H}om_E(F, F)) \subseteq \text{supp}(\mu\mathcal{H}om_{\mathcal{O}_E/\pi}(F_1, F_1)),$$

which is exactly (8.2.6) after projecting away the suppressed η_0 -factor.

Finally, F_1 is a constructible complex with coefficients in the finite field $\mathcal{O}_E/\pi$. Under the equivalence between constructible torsion sheaves on the pro-étale and the étale sites, Theorem 4.2 applies to F_1 and gives $\text{SS}_\mu(F_1) \subseteq \text{SS}(F_1)$. Together with (8.2.6) and (8.2.3), we obtain

$$(8.2.11) \quad \text{SS}_\mu(F) \subseteq \text{SS}_\mu(F_1) \subseteq \text{SS}(F_1) = \text{SS}(F),$$

as required. $\square$

Within the six-functor formalism, the microlocal characteristic cycle CC_μ for an object $F \in D_{\text{cons}}(X_{\text{proét}}, E)$ is also well-defined.

Proposition 8.3 ([Sai26, Proposition 2.3.3 and Proposition 2.5.1]). *Let $f : X \rightarrow Y$ be a proper morphism of smooth schemes over k and $F \in D_{\text{cons}}(X_{\text{proét}}, E)$. Then we have*

$$(8.3.1) \quad \text{CC}_\mu(Rf_*F) = f_!\text{CC}_\mu(F) \quad \text{in} \quad H_{f_*\text{SS}_\mu(F) \cup \text{SS}_\mu(Rf_*F)}^0(T^*Y, e_Y^{\vee*}\mathcal{K}_{Y/k}),$$

where $e_Y^{\vee} : T^*Y \rightarrow Y$ is the projection.

In particular if X is a smooth curve and F is smooth outside a finite set $Z \subseteq X$, then we have

$$(8.3.2) \quad \text{CC}_\mu(F) = -\text{rank} F \cdot \text{cl}[T_X^*X] - \sum_{x \in Z} a_x(F) \cdot \text{cl}[T_x^*X],$$

where $a_x(F)$ is the Artin conductor of F at x .

Proof. The proof of [Sai26, Proposition 2.3.3 and Proposition 2.5.1] extend to the pro-étale setting within the six-functor formalism. Here are the details.

Consider the following correspondence

$$(8.3.3) \quad T^*X \xleftarrow{q=df} X \times_Y T^*Y \xrightarrow{p} T^*Y.$$

Thus p is proper and $f_*C = p(q^{-1}C)$ for a closed conical subset $C \subset T^*X$. The microlocal characteristic cycles are defined by the compositions

$$(8.3.4) \quad E_{T^*X} \xrightarrow{u_X} \mu\mathcal{H}om_E(F, F) \xrightarrow{v_X} e_X^{\vee*}\mathcal{K}_{X/k}, \quad E_{T^*Y} \xrightarrow{u_Y} \mu\mathcal{H}om_E(Rf_*F, Rf_*F) \xrightarrow{v_Y} e_Y^{\vee*}\mathcal{K}_{Y/k},$$

where u_X and u_Y are the identities, v_X and v_Y evaluation morphisms. Now we use the proof of [Sai26, Proposition 2.3.3]. Consider the following commutative diagram (cf. [Sai26, (2.53)])

$$(8.3.5) \quad \begin{array}{ccccc} X & \xlongequal{\quad} & X & \xrightarrow{f} & Y \\ \Delta_X \downarrow & & \Gamma_f \downarrow & & \downarrow \Delta_Y \\ X \times X & \xrightarrow{\text{id}_X \times f} & X \times Y & \xrightarrow{f \times \text{id}_Y} & Y \times Y \end{array}$$

Consider the middle horizontal arrows in [Sai26, (2.67) and (2.68)], they define a canonical morphism

$$(8.3.6) \quad \beta_f : Rp_*q^*\mu\mathcal{H}om_E(F, F) \longrightarrow \mu\mathcal{H}om_E(Rf_*F, Rf_*F).$$

Let $\eta_p : E_{T^*Y} \rightarrow Rp_*E_W$ be the unit. Smooth base change for $!$ -pullback gives $q^*e_X^{\vee*}\mathcal{K}_{X/k} \simeq p^!e_Y^{\vee*}\mathcal{K}_{Y/k}$, and hence a trace morphism $\tau_p : Rp_*q^*e_X^{\vee*}\mathcal{K}_{X/k} \longrightarrow e_Y^{\vee*}\mathcal{K}_{Y/k}$. The unit, counit and evaluation identities give

$$(8.3.7) \quad \beta_f \circ Rp_*(q^*u_X) \circ \eta_p = u_Y, \quad v_Y \circ \beta_f = \tau_p \circ Rp_*(q^*v_X).$$

We justify (8.3.7) for E -coefficients by reduction. Choose an integral model F_0 , put $F_m = F_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m$. Since f is proper, we have an isomorphism $(Rf_*F_0) \otimes_{\mathcal{O}_E}^L A_m \simeq Rf_*F_m$. Verdier duality, external products, specialization, Fourier–Deligne transformation, and their unit, counit, evaluation and exchange morphisms commute with these coefficient changes. Consequently the finite-coefficient diagrams of [Sai26, Proposition 2.3.3] form a compatible inverse system. The equivalence

$$(8.3.8) \quad D_{\text{cons}}(-, \mathcal{O}_E) \simeq \varprojlim_m D_{\text{ctf}}^b(-, \mathcal{O}_E/\pi^m)$$

shows that the corresponding diagram commutes over $\mathcal{O}_E$; rationalizing gives (8.3.7).

Now put $C = f_* \text{SS}_\mu(F) \cup \text{SS}_\mu(Rf_*F)$. The two middle terms in (8.3.7) are supported on C . Applying the supported-trace factorization defining the microlocal characteristic class gives

$$(8.3.9) \quad f_! \text{CC}_\mu(F) = \text{CC}_\mu(Rf_*F) \quad \text{in} \quad H_C^0(T^*Y, e_Y^{\vee*} \mathcal{K}_{Y/k}).$$

This proves (8.3.1).

Now assume X is a smooth curve. We prove (8.3.2). Put $D = T_X^*X \cup \bigcup_{x \in Z} T_x^*X$. Choose an integral model F_0 which is lisse over $X \setminus Z$. Semi-purity and derived π -completeness give

$$(8.3.10) \quad H_D^0(T^*X, e_X^{\vee*} \mathcal{K}_{X/k, \mathcal{O}_E}) \xrightarrow{\sim} \varprojlim_m H_D^0(T^*X, e_X^{\vee*} \mathcal{K}_{X/k, \mathcal{O}_E/\pi^m}).$$

Indeed, the relevant supported complexes lie in $D^{\geq 0}$, so the possible $\varprojlim^1 H^{-1}$ -term vanishes.

For every m , by [Sai26, Proposition 2.5.1] we have

$$(8.3.11) \quad \text{CC}_\mu(F_m) = -\text{rank}(F) \text{cl}_{\mathcal{O}_E/\pi^m}[T_X^*X] - \sum_{x \in Z} a_x(F) \text{cl}_{\mathcal{O}_E/\pi^m}[T_x^*X].$$

Here rank and conductor are unchanged by derived coefficient reduction. More precisely, vanishing cycles commute with reduction, and we have an isomorphism

$$(8.3.12) \quad R\Phi_x(F_0) \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m \simeq R\Phi_x(F_m);$$

Since the alternating rank of these perfect complexes is unchanged. Thus $a_x(F_m) = a_x(F)$ (see also [UYZ20, (5.3.4.1)]). Passing to the inverse limit using (8.3.10), and then inverting π , proves

$$(8.3.13) \quad \text{CC}_\mu(F) = -\text{rank}(F) \text{cl}[T_X^*X] - \sum_{x \in Z} a_x(F) \text{cl}[T_x^*X].$$

□

Corollary 8.4. *Let X be a smooth connected k -scheme, and let $F \in D_{\text{cons}}(X_{\text{proét}}, E)$. Then we have*

$$(8.4.1) \quad \text{CC}_\mu(F) = \text{cl}(\text{CC}(F)),$$

where $\text{cl} : Z_{\dim(X)}(\text{SS}(F)) \rightarrow H_{\text{SS}(F)}^0(T^*X, \pi^* \mathcal{K}_{X/k})$ is the cycle class map.

Proof. By [Sai26, Proposition 2.1.7] and [Sai17, Lemma 5.13.1], both CC_μ and CC are additive for distinguished triangles. By [Bar24, Theorem 1.5(viii)], we have

$$(8.4.2) \quad \text{SS}(F) = \bigcup_i \text{SS}({}^p H^i F).$$

By passing to the perverse cohomology sheaves and then to their Jordan–Hölder constituents, we may assume that F is a perverse E -sheaf.

Take a torsion-free perverse integral model F_0 . Put $F_1 = F_0 \otimes^L \mathcal{O}_E/\pi$. By [Bar24, Theorem 1.5(v)], we have

$$(8.4.3) \quad \text{SS}(F_1) = \text{SS}(F)$$

for every such model. By definition,

$$(8.4.4) \quad \mathrm{CC}(F_1) = \mathrm{CC}(F).$$

We may assume $F \neq 0$. Now we follow the proof of [Sai26, Proposition 2.5.2]. Let $C = \mathrm{SS}(F_1) = \mathrm{SS}(F)$. By [Bei16, Theorem 1.3], we have $\dim C = \dim X$. By semi-purity, we have $H_C^0(T^*X, e^{\vee*}\mathcal{K}_X) = \oplus_a E[C_a]$, where C_a runs through irreducible components of C (of dimension $\dim X$). Hence $\mathrm{CC}_\mu(F) = \sum_a \mu_a[C_a]$ for some $\mu_a \in E$. Let $CC = \sum_a m_a[C_a]$. It suffices to show the equality $\mu_a = m_a$ in E for each irreducible component C_a .

Since the question is local on X , we may assume that X is affine. By applying push-forward to an immersion $X \rightarrow \mathbb{A}^n$, we may assume that $X = \mathbb{A}^n$. Further, we may assume that X is projective. Now we fix a closed immersion $X \rightarrow \mathbb{P}^N$ for N large enough.

Let C_a be an irreducible component of C . Then, there exists a linear subvariety $A \subset \mathbb{P}^N$ of codimension 2 and $x \in X \setminus X \cap A$ satisfying the following conditions. The intersection $X \cap A$ is transverse. Let $\pi : X' \rightarrow X$ be the blow-up along $X \cap A$ and let $f : X' \rightarrow Y = \mathbb{P}^1$ be the morphism to the projective line parametrizing hyperplanes containing A . The fibers of f are the intersections with hyperplanes. The point x is an isolated characteristic point of $f : X' \rightarrow Y$, and x is the unique characteristic point in the fiber $f^{-1}(y)$ for $y = f(x)$. The component $C_a \subset T^*X$ is the unique irreducible component of C meeting the section df at x . The intersection number $(C_a, df)_x$ equals 1, if necessary replacing X by $X \times \mathbb{P}^1$ in the case $p = 2$ [Sai17, Proposition 5.19].

Let m_a be the coefficient of C_a in $\mathrm{CC}\pi^*F = \mathrm{CC}\pi^*F_1$, and let $\mu_a \in E$ be the coefficient of $[C_a]$ in $\mathrm{CC}_\mu F$. Then, we have $(\mathrm{CC}\pi^*F_1, df)_x = m_a(C_a, df)_x \in \mathbb{Z}$ and $(\mathrm{CC}_\mu\pi^*F, df)_x = \mu_a(C_a, df)_x \in E$. By the Milnor formula [Sai17, Theorem 5.9], we have

$$(8.4.5) \quad \dim \mathrm{tot} R\Phi_x(\pi^*F_1, f) = -(\mathrm{CC}\pi^*F_1, df)_x = -m_a(C_a, df)_x.$$

By Proposition 8.3, we have $f_!CC_\mu\pi^*F = CC_\mu Rf_*\pi^*F$. Thus the intersection product $(CC_\mu F, df)_x = \mu_a(C_a, df)_x$ equals the coefficient $-a_y(f_*\pi^*F)$ of $[T_y^*Y]$ in $f_!CC_\mu\pi^*F = CC_\mu f_*\pi^*F$. By [UYZ20, (5.3.4.1)], we have

$$(8.4.6) \quad a_y(f_*\pi^*F_1) = \dim \mathrm{tot} R\Phi_x(\pi^*F_1, f) = \dim \mathrm{tot} R\Phi_x(\pi^*F, f) = a_y(f_*\pi^*F).$$

Thus, we have

$$(8.4.7) \quad \dim \mathrm{tot} R\Phi_x(\pi^*F_1, f) = -m_a(C_a, df)_x = -\mu_a(C_a, df)_x.$$

Since $(C_a, df)_x = 1$, we obtain $m_a = \mu_a$ in E , and hence $CC_\mu F = \mathrm{cl} CCF$. $\square$

8.5. Let $i : X \rightarrow \mathbb{P} = \mathbf{P}(W^\vee)$ be an immersion, and let $C_a \subset \mathrm{SS}(\mathcal{P})$ be an irreducible component. Let $\tilde{C}_a \subset X \times_{\mathbb{P}} T^*\mathbb{P}$ be its inverse image. For a line $L \subset \mathbb{P}^\vee$, the axis $A_L \subseteq \mathbb{P}$ is a linear subspace of codimension 2 defined as the intersection of the hyperplanes parametrized by L . Put

$$(8.5.1) \quad Q_L = \{(u, H') \in \mathbb{P} \times L \mid u \in H'\}, \quad p_L : Q_L \rightarrow \mathbb{P}, \quad g_L : Q_L \rightarrow L.$$

Then Q_L is the blow-up of $\mathbb{P}$ along A_L , and g_L is proper.

Theorem 8.6. *For every perverse E -sheaf $\mathcal{P}$ on a smooth k -scheme X , we have*

$$(8.6.1) \quad \mathrm{SS}_\mu(\mathcal{P}) = \mathrm{SS}(\mathcal{P}).$$

Proof. The assertion may be checked on the connected components of X . We may therefore assume that X is connected, and put $n = \dim X$. The assertion is clear if $\mathcal{P} = 0$. By Theorem 8.2, we have

$$(8.6.2) \quad \mathrm{SS}_\mu(\mathcal{P}) \subseteq \mathrm{SS}(\mathcal{P}).$$

We first establish the positivity property of $\mathrm{CC}(\mathcal{P})$. By [Bar24, Proposition 2.6 and the proof of Theorem 1.5(v)], we may choose an integral model $\mathcal{P}_0 \in \mathrm{Perv}(X, \mathcal{O}_E)$ of $\mathcal{P}$ which is torsion-free as a perverse sheaf. Put $\mathcal{P}_1 = \mathcal{P}_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi$. The distinguished triangle

$$(8.6.3) \quad \mathcal{P}_0 \xrightarrow{\pi} \mathcal{P}_0 \longrightarrow \mathcal{P}_1 \longrightarrow$$

shows that $\mathcal{P}_1$ is perverse. Indeed, ${}^p H^{-1}(\mathcal{P}_1) = \ker(\pi : \mathcal{P}_0 \rightarrow \mathcal{P}_0) = 0$ because $\mathcal{P}_0$ is torsion-free, while ${}^p H^0(\mathcal{P}_1) = \operatorname{coker}(\pi : \mathcal{P}_0 \rightarrow \mathcal{P}_0)$. By [Bar24, Theorem 1.5(ii),(v)], we have

$$(8.6.4) \quad \operatorname{SS}(\mathcal{P}_1) = \operatorname{SS}(\mathcal{P}_0) = \operatorname{SS}(\mathcal{P}).$$

By [Sai17, Proposition 4.14], the characteristic cycle $\operatorname{CC}(\mathcal{P}_1)$ is effective and

$$(8.6.5) \quad \operatorname{supp} \operatorname{CC}(\mathcal{P}_1) = \operatorname{SS}(\mathcal{P}_1).$$

Since the characteristic cycle with E -coefficients is defined by reduction of an integral model, if $\operatorname{SS}(\mathcal{P}) = \bigcup_{a \in A} C_a$ is the decomposition into irreducible components, then

$$(8.6.6) \quad \operatorname{CC}(\mathcal{P}) = \operatorname{CC}(\mathcal{P}_1) = \sum_{a \in A} m_a [C_a], \quad m_a \in \mathbf{Z}_{>0}.$$

In particular, $\operatorname{SS}(\mathcal{P})$ is purely n -dimensional, and every m_a has nonzero image in the characteristic-zero field E .

We now recall the consequence of semi-purity needed below. If $W \subseteq T^*X$ is a closed subset of dimension at most n , then the cycle-class map induces an isomorphism

$$(8.6.7) \quad \operatorname{cl}_{W,E} : Z_n(W) \otimes_{\mathbf{Z}} E \xrightarrow{\sim} H_W^0(T^*X, e_X^{\vee*} \mathcal{K}_{X/k}),$$

where $Z_n(W)$ denotes the group of n -dimensional cycles supported on W .

Indeed, since X is smooth of dimension n , we have $e^* \mathcal{K}_{X/k} \simeq E(n)[2n]$. A closed subset of W of dimension strictly smaller than n has codimension at least $n+1$ in the smooth scheme T^*X of dimension $2n$. Semi-purity therefore shows that such a subset contributes neither to degree 0 nor to degree 1 cohomology with coefficients in $e^* \mathcal{K}_{X/k}$. After removing a closed subset of dimension less than n , the n -dimensional irreducible components of W become pairwise disjoint regular closed subschemes of codimension n . Absolute purity then gives one copy of E for each such component, proving (8.6.7). Moreover, since these identifications are induced by cycle classes, they are compatible with extension of supports.

Let $C = \operatorname{SS}(\mathcal{P})$, $C_\mu = \operatorname{SS}_\mu(\mathcal{P})$ and

$$\rho_{C_\mu, C} : H_{C_\mu}^0(T^*X, e^* \mathcal{K}_{X/k}) \longrightarrow H_C^0(T^*X, e^* \mathcal{K}_{X/k})$$

be the extension-of-supports morphism. Since $C_\mu \subseteq C$ and C is purely n -dimensional, every n -dimensional irreducible component of C_μ is one of the C_a . Consequently, (8.6.7) gives

$$(8.6.8) \quad \operatorname{Im}(\rho_{C_\mu, C}) = \bigoplus_{\{a \in A : C_a \subseteq C_\mu\}} E \cdot \operatorname{cl}_{C,E}([C_a]) \subseteq H_C^0(T^*X, e^* \mathcal{K}_{X/k}).$$

By Corollary 8.4, interpreted after extension of supports along $C_\mu \subseteq C$, we have

$$(8.6.9) \quad \begin{aligned} \rho_{C_\mu, C}(\operatorname{CC}_\mu(\mathcal{P})) &= \operatorname{cl}_{C,E}(\operatorname{CC}(\mathcal{P})) \\ &= \sum_{a \in A} m_a \operatorname{cl}_{C,E}([C_a]) \quad \text{in} \quad H_C^0(T^*X, e^* \mathcal{K}_{X/k}). \end{aligned}$$

The left-hand side of (8.6.9) belongs to the subspace described in (8.6.8). On the right-hand side, however, every coefficient m_a is nonzero in E . It follows that $C_a \subseteq C_\mu$ for every $a \in A$. Therefore $C \subseteq C_\mu$. Together with (8.6.2), this proves $\operatorname{SS}_\mu(\mathcal{P}) = \operatorname{SS}(\mathcal{P})$. $\square$

APPENDIX A. COHOMOLOGICAL CORRESPONDENCES

In this section, we recall the theory of cohomological correspondences.

A.1. Let B be a base scheme and Sch_B the category of schemes separated and of finite type over B . We have the symmetric monoidal category $\text{Corr}(\text{Sch}_B)$ of correspondences over B . We denote by $[g_! f^*]$ a correspondence $(X \xleftarrow{f} Z \xrightarrow{g} Y)$ from X to Y . Consider the six-functor formalism

$$(A.1.1) \quad D_{\text{ctf}}^b : \text{Corr}(\text{Sch}_B) \rightarrow \text{Cat}, X \mapsto D_{\text{ctf}}^b(X, \Lambda).$$

The symmetric monoidal category CohCorr_B is the unstraightening of (A.1.1). We also need the category $\text{CohCorr}_B^\otimes$ of operations in CohCorr_B (cf. [Lur17, Construction 2.1.1.7]).

- (1) An object of CohCorr_B is a pair $(X; F)$ where $X \in \text{Sch}_B$ and $F \in D_{\text{ctf}}^b(X, \Lambda)$.
- (2) A morphism $([g_! f^*]; \alpha) : (X; F) \rightarrow (Y; G)$ in CohCorr_B consists of a correspondence

$$(A.1.2) \quad X \xleftarrow{f} Z \xrightarrow{g} Y \quad \text{in } \text{Sch}_B,$$

together with a cohomological correspondence

$$(A.1.3) \quad \alpha : f^* F \rightarrow g^! G.$$

- (3) The tensor product in CohCorr_B is

$$(A.1.4) \quad (X; F) \otimes_B (Y; G) = (X \times_B Y; F \boxtimes_B G).$$

- (4) A multi-morphism $(X_1, \dots, X_n; F_1, \dots, F_n) \rightarrow (Y; G)$ in $\text{CohCorr}_B^\otimes$ is a morphism

$$(A.1.5) \quad (X_1; F_1) \otimes_B \dots \otimes_B (X_n; F_n) \rightarrow (Y; G).$$

For $n = 0$, we denote by $[\Lambda] : () \rightarrow (Y; \Lambda)$ the cocartesian 0-ary morphism.

A.2. Let S be a strictly henselian trait. Let $s \in S$ be the geometric closed point and $\eta \in S$ be the geometric generic point. By [LZ22, Construction 3.3], we have two symmetric monoidal categories $\text{CohCorr}_{S,s}$, $\text{CohCorr}_{S,\eta}$ and a symmetric monoidal functor

$$(A.2.1) \quad \Psi : \text{CohCorr}_{S,\eta} \rightarrow \text{CohCorr}_{S,s},$$

which we describe here:

- (1) $\text{CohCorr}_{S,s}$ is the unstraightening of the lax symmetric monoidal functor

$$(A.2.2) \quad \text{Corr}(\text{Sch}_S) \xrightarrow{(\cdot) \times_{S^s}} \text{Corr}(\text{Sch}_s) \xrightarrow{D_{\text{ctf}}^b} \text{Cat}, X \mapsto D_{\text{ctf}}^b(X_s, \Lambda).$$

Following the notation for CohCorr_S , we use the subscript s for objects and morphisms in this category. We suppress the

- (2) $\text{CohCorr}_{S,\eta}$ is the unstraightening of the lax symmetric monoidal functor

$$(A.2.3) \quad \text{Corr}(\text{Sch}_S) \xrightarrow{(\cdot) \times_{S^\eta}} \text{Corr}(\text{Sch}_\eta) \xrightarrow{D_{\text{ctf}}^b} \text{Cat}, X \mapsto D_{\text{ctf}}^b(X_\eta, \Lambda).$$

Following the notations for CohCorr_S , we use the subscript η for objects and morphisms in this category.

- (3) The functor Ψ sends $(X; F)_\eta$ to $(X; \Psi(F))_s$.

For convenience, we omit the Galois action in the notation.

A.3. Since Ψ is a symmetric monoidal functor, it induces a functor

$$(A.3.1) \quad \Psi^\otimes : \text{CohCorr}_{S,\eta}^\otimes \rightarrow \text{CohCorr}_{S,s}^\otimes.$$

This functor encodes the coherent compatibility of nearby cycles with the three functors: $*$ -pullback, $!$ -pushforward and tensor product.

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